

Heterogeneous Agents In Developed and Developing Countries

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I study whether an economy's response to aggregate shocks depends on its level of development. Empirically, I estimate responses to high-frequency monetary shocks at the macro and micro level and compute hand-to-mouth shares for high, middle and low income countries. An inverted U-shaped relationship between these two is found. Theoretically, I show that matching this relationship requires a heterogeneous agent model with one key feature: countercyclical inequality and income risk. This generates amplifying income effects by shifting towards constrained households the additional aggregate demand generated by unconstrained households' substitution effects. Amplification peaks with heterogeneity, when there is an equal share of households on and off their Euler equations. These results have implications for aggregate demand stabilization policies and the welfare costs of business cycles.

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1 Introduction

How do economies at different stages of development respond to aggregate shocks? Addressing this question would help us understand whether business cycles are different between developed and developing countries, and whether we are in need of country-specific aggregate demand stabilization policies. However, our understanding of this cross-country relationship is limited, with most estimates coming from high-income economies. This should come as no surprise, as such enterprise faces not one, but two empirical challenges: the lack of exogenous shocks in macroeconomics and the fact that data availability is endogenous to development.

Neither there is a conclusive theoretical answer to such question, despite borrowing constraints being pervasive in developing countries and the distinctive feature of incomplete markets models. One illustrative case comes from monetary policy. Partial equilibrium logic would point to a larger share of the population being excluded from credit markets, stripping interest rates from their ability to stimulate aggregate demand. But general equilibrium forces could be unleashed if this additional aggregate demand relaxes borrowing constraints, pushing output in the opposite direction. These effects might even cancel each other out, making the incomplete markets structure of second order.

To shed light on the empirical relationship and distinguish between those theoretical predictions, I introduce a new moment that consists of an inverted U-shaped relationship between output responses to monetary shocks and the hand-to-mouth share of households, across 27 countries. Consider three of them: Italy with a hand-to-mouth share around 15%, Cyprus with 45%, and Thailand with nearly 75%. I find that output's reaction to the same monetary shock is nearly three times as large in Cyprus as in Italy, while it is negligible in Thailand. Micro estimates replicate the fact. The heterogeneity in response to shocks favors a heterogeneous agent New Keynesian (HANK) model over a representative agent (RANK) one. In particular, countercyclical inequality and income risk in a one-asset HANK model is required to match this moment.

I try to address the dual empirical challenge outlined above in section 2. To deal with the endogeneity of macroeconomic shocks, I estimate country-specific Proxy SVAR output impulse response functions to high frequency identified monetary shocks. I exploit two types of external instruments: an interest rate based one coming from shocks in futures interest rates markets and an exchange rate one based on shocks to forward premiums. They are used to estimate aggregate shocks responses measures. To overcome the data constraints, I assemble a household level dataset with information on assets, income and consumption coming from surveys of countries across the development spectrum like Italy and Thailand. Hand-to-mouth shares are computed out of this data, using [Kaplan et al. \(2014\)](#)'s methodology . Such is the measure of development. Pulling together these two measures gives way to the inverted U-shaped relationship. The moment gets confirmed at the micro level, since the causal responses to monetary shocks can also be estimated at the household level exploiting the panel structure of the microdata. The effects statistically differ by household's hand-to-mouth status.

I discipline the theoretical predictions with this novel moment in section 3. A RANK model fails to account for such a fact since it predicts that the output response will be the

same independently of the hand-to-mouth share. Instead, I demonstrate that a one-asset HANK model with countercyclicality of inequality and income risk can match the inverted U-shape. I then bring in fiscal policy and investment and find that, as long as they are not introduced in a countercyclical way, they cannot explain the empirical fact. Along the way I provide micro data validation for all of this: the one-asset HANK model also matches marginal propensities to consume (MPC) estimates coming from Havranek and Sokolova (2020) and countercyclicality is found at the individual level.

How cyclicity is specified turns out to be critical since it determines whether shocks get amplified by borrowing constraints. It does so by controlling how the income distribution moves with aggregate income, with an expansive monetary shock concentrating agents' incomes under the countercyclical case. This relaxes the borrowing constraints of low-income, high-MPC households by redistributing towards them the extra aggregate demand (countercyclical inequality) and reduces their savings due to precautionary motives (countercyclical income risk). Both mechanisms amplify through constrained households' income effects the initial unconstrained households' substitution effects.

The empirical moment's inverted U-shape is a result of the interaction between such amplification mechanism and how the hand-to-mouth share of agents changes across the development spectrum. In high-income countries, the majority of the population can intertemporally substitute. This gets modestly amplified by the income effect that countercyclicality generates when redistributing towards the modest amount of constrained households. In middle-income countries, the share of agents on their Euler equation is smaller, but still enough to have a significant direct impact on aggregate demand. Given that the share of constrained agents is larger, this triggers an even stronger indirect effect. In low-income countries, there are so few agents who have access to credit markets that the initial partial equilibrium response is not meaningful. Hence, there are no large general equilibrium effects despite the large mass of constrained agents.

Coming back to the initial question posed, this paper argues that aggregates' reaction to shocks do differ along the development path. Though I provide an answer in the context of monetary shocks, this has broader business cycle implications. Similar mechanisms (that not necessarily transmit into the same response) could be at play for productivity and fiscal policy shocks. Optimal aggregate demand stabilization policies will also need to be different across countries as a consequence. The higher potency of monetary policy in middle-income countries could be a blessing, but also a curse if not administered correctly. In terms of welfare, the incomplete market structure by itself augments the costs of business cycles. They might be even larger in emerging markets if, as this paper tries to show, a higher share of hand-to-mouth agents makes aggregate demand more volatile.

Literature. The present paper relates to the large literature that investigates business cycles through the lens of heterogeneous agents models.¹ The empirical support of these papers comes from developed countries evidence. This paper instead picks its moments from the cross-section of high, middle and low-income countries. A theoretical exception is Werning (2015), who finds conditions under which aggregates in incomplete mar-

¹ A non-comprehensive list is McKay et al. (2016), Auclert et al. (2018), Auclert (2019), Wong et al. (2019), Auclert et al. (2020), Ottonello and Winberry (2020), Auclert et al. (2021b) and Guerreiro (2023).

kets models react as in a representative agent model for any level of liquidity.² This neutrality result is line with Krusell and Smith (1998) and finds empirical support in Berger et al. (2023), who show that deviations from complete markets account for only 7% of U.S. output volatility. In comparison, my evidence suggests that the degree of market incompleteness matters. The different conclusions with respect to those papers stem from the complementarity between borrowing countries and countercyclicality that leads to considerable amplification in middle-income countries. Another strand of the literature that Werning (2015) speaks to, sheds light into HANK mechanisms by imposing assumptions that allow them to derive analytical results (see Acharya and Dogra 2020, Ravn and Sterk 2021, Bilbiie 2024). They find that cyclical inequality and income risk are the key parameters that determine amplification. This paper aims to better understand the mechanisms at play by taking an empirical route instead of aiming for tractability.

A converse set of papers explore macroeconomic fluctuations across countries, but abstaining from modeling any kind of heterogeneity. The present paper builds on Almgren et al. (2022), who study interest rate shocks responses within a monetary union and on Witheridge (2024), who studies monetary-fiscal interactions in emerging markets. Which brings us to the large literature coming out from Neumeyer and Perri (2005) and Aguiar and Gopinath (2007) that revolves around the excessive volatility of consumption and investment that these economies exhibit.³ Relative to the middle-income economies that are their object of study, this paper brings in country-specific estimates for both low and high income economies, with hand-to-mouth shares that range from around 15% for Italy to 75% for Thailand. Pulling together all of them under the same methodology, I show that emerging markets are a special case where the HANK mechanisms are at play in their most extreme form. Moreover, this paper suggests that the incomplete markets structure may offer a complementary explanation to the excess volatility puzzle, similar to the role financial frictions play in Garcia-Cicco et al. (2010) and Mendoza (2010).

The intersection between these literatures are papers that empirically exploit cross-sectional variation and theoretically model heterogeneity. Herreño and Pedemonte (2022)'s TANK model generates a positive relation between employment and prices after a monetary shock. Moreover, they find that low-income U.S. regions have a higher sensitivity. Hong (2020) can explain emerging markets' excess volatility by feeding *transitory* productivity and financial shocks to a small open economy HANK model. This is true when the model is calibrated to match Peruvian MPCs from Hong (2023), but not when calibrated to U.S. MPCs. Conversely, Guntin et al. (2023) find that *permanent* TFP shocks to their small open economy heterogeneous agents model can rationalize the large consumption adjustments that they observe along the income distribution. This occurs after large recessions in Italy, México, Perú and Spain.⁴ Relative to those papers, empirically I add cross-country variation that turns out to be crucial to uncover the inverted U-shape between aggregates' responses to *monetary* shocks and development. Theoretically, I isolate countercyclical inequality and income risk as the key mechanism that makes the incomplete market structure have amplifying effects.⁵

² The key assumption needed is acyclicity of income risk. Werning (2015) also studies deviations from these assumptions and shows how countercyclical income risk can amplify monetary shocks.

³ A non-exhaustive list is Arellano (2008), Schmitt-Grohé and Uribe (2016) and Casal and Guntin (2023).

⁴ Hong (2020) and Guntin et al. (2023) findings can be reconciled if one thinks of garden-variety recessions as transitory TFP shocks and crises as permanent TFP shocks.

⁵ Herreño and Pedemonte (2022) and Hong (2020) get amplification by implicitly assuming counter-

2 Empirics

In this empirical section I first present the data and methodologies to measure both sides of the relationship between aggregate responses to shocks and development. I display the estimates of such methods and the resulting relationship afterwards. Lastly, I discuss alternative methods and how robust results are.

2.1 Macro Evidence: Baseline Methods

In order to uncover the relationship between aggregate shocks and development, I need to proceed in two steps. In the first step I set the baseline Proxy SVAR empirical strategy used to estimate output responses to monetary shocks. In the second step I outline how to calculate hand-to-mouth shares, my baseline measure of borrowing constraints.

Measuring Output Responses to Monetary Shocks

In this first step, I measure cross-country aggregates responses to shocks as output reactions to monetary shocks at the country-level. I start by describing the aggregate data I utilize, then proceed to present the Proxy SVAR framework used as empirical strategy and finally describe the two families of high frequency instruments utilized for identification.

Data. For the SVAR, I assemble a panel of 27 countries with variables at a monthly frequency. The sample of countries consists of Austria, Belgium, China, Croatia, Cyprus, Czechia, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Mexico, Netherlands, Portugal, Russia, Slovakia, Slovenia, South Africa, Spain and Thailand. Variables included are interest rate, unemployment rate, seasonally adjusted industrial production in logs and CPI in logs, which mostly come from *IMF IFS*. Exceptions are listed in [Appendix A](#). The period covered in the sample spans the months from January 2010 to December 2019.

For the proxies, I also create a panel, that is composed of two types of instruments. The first type employs one-year interest rate swaps, where the index rate is the overnight rate, from [Bolhuis et al. \(2024\)](#). The second type relies on exchange rates, calculated using spot and 12-month forward exchange rates, following [Witheridge \(2024\)](#). Central bank policy meeting dates are also needed and they come from *Bloomberg Calendar*, *Central Bank News* and from central banks themselves. Details can be found in [Appendix B](#). The frequency is monthly and the period covered is January 2010 - December 2019.⁶

Data limitations prevent us from having both the interest rate and the exchange rate based instruments available for all 27 countries. However, overlap is high: 21 countries have both proxies. The interest rate based instrument is not available for South Africa. The exchange rate based instrument is not available for China, Czechia, Hungary, Mexico and Thailand.

cyclicality. The former through labor supply, the latter by introducing financial frictions shocks.

⁶ Nothing, theoretically, prevents the variables included in the VAR to cover a larger sample than the proxy. In fact, this would produce more precise estimates. However, for some countries, interest rates are not available going further back than 2009. This determines the lower bound of the sample. The upper bound is set by the start of the COVID-19 pandemic.

Proxy SVAR. Separately for each country, consider the reduced-form order 4 VAR

$$\mathbf{y}_t = \mathbf{b} + \sum_{l=1}^3 \mathbf{B}_l \mathbf{y}_{t-l} + \mathbf{u}_t,$$

where $\mathbf{y}_t$ is a vector of endogenous variables. In particular, $\mathbf{y}_t = [r_t, ip_t, u_t, p_t]$, where r_t is monthly interest rate, ip_t is monthly seasonally adjusted industrial production in logs, u_t is unemployment rate and p_t is monthly CPI in logs. $\mathbf{b}$ is a vector of constants, $\mathbf{B}_1, \dots, \mathbf{B}_L$ are matrices of coefficients and $\mathbf{u}_t$ is a vector of residuals with covariance matrix σ . $\mathbf{u}_t$ is the linear transformation of the vector of structural shocks $\epsilon_t = [\epsilon_{r,t}, \epsilon_{ip,t}, \epsilon_{u,t}, \epsilon_{p,t}]$, which have covariance matrix $\mathbf{I}$, generated by the matrix $\mathbf{S}$

$$\mathbf{u}_t = \mathbf{S}\epsilon_t,$$

with $\mathbf{S}$ coming from $\mathbf{S}\mathbf{S}' = \Sigma$. This is satisfied by infinitely many matrices $\mathbf{S}$, which is the usual identification challenge from VAR. A well-known approach is to proceed with a Cholesky decomposition, but it is also well-known that it implies strong assumptions.

An alternative way path are external instruments. For identification to be successful, the first column of $\mathbf{S}$, $\mathbf{s}_1$, needs to be identified. This corresponds to the variable to be shocked in the SVAR, namely, r_t . Therefore, if one has a proxy $\mathbf{z}$ that satisfies

$$\begin{aligned} \mathbb{E}[z_t \epsilon_{r,t}] &\neq 0 \\ \mathbb{E}[z_t \epsilon_{ip,t}] &= \mathbb{E}[z_t \epsilon_{u,t}] = \mathbb{E}[z_t \epsilon_{p,t}] = 0, \end{aligned}$$

where $\epsilon_{r,t}$ is the shock corresponding to $\mathbf{s}_1$, and $\epsilon_{2:n,t}$ represents the other structural shocks, one would achieve identification of $\mathbf{s}_1$ up to sign and scale.⁷

External Instrument. I use two classes of country-specific high-frequency monetary shocks.

The first class is based on interest rates, following Bolhuis et al. (2024). During a narrow window around the country's central bank monetary policy announcement, the high frequency shock is

$$\Delta r_t = \mathbb{E}_t r_t - \mathbb{E}_{t-1} r_t,$$

where Δr_t is the change in the interest rate during such window and $\mathbb{E}_t r_t$ denotes the market expectation of the interest rate. Hence, the high frequency shock is the revision in expected rates that occurs during the announcement event. A well-known concern is that part of this surprise may reflect the central bank's private information about the economy rather than an exogenous policy action.⁸ To address this, one regresses

$$\Delta r_t = \alpha + \beta \mathbf{X}_{t-1} + \varepsilon_t,$$

where α is a constant, $\mathbf{X}_{t-1}$ is a vector of controls and ε_t is the residual, which is kept as the *orthogonalized* high frequency monetary policy shock. This captures pure policy rate changes uncorrelated with information publicly available prior to the announcement.

⁷ The scale was determined when setting the covariance matrix of structural shocks ϵ_t equal to $\mathbf{I}$, giving way to the following interpretation: an additional unit of $\epsilon_{1,t}$ has an effect of one standard deviation on $\mathbf{y}_{1,t}$.

⁸ See, e.g., Miranda-Agrippino and Ricco (2021), Jarociński and Karadi (2020), Bauer and Swanson (2023a,b). Appendix C provides details.

However, though present in certain emerging economies, interest rate futures markets are not widespread in low income countries. But exchange rate futures markets are. Therefore, in order to get higher coverage, the second class constructs high frequency monetary shocks based on exchange rates, as in Witheridge (2024). This is achieved by exploiting the country's Covered Interest Parity (CIP)

$$f_t - e_t = r_t - r_t^*,$$

where f_t is the log of the 12-month forward exchange rate against the U.S. dollar, e_t is the log of the spot exchange rate, r_t is the country's interest rate and r_t^* is the U.S. interest rate. During the narrow window in which the monetary policy announcement takes place, r_t^* remains plausibly constant. Then, high frequency monetary shocks can be measured as

$$\Delta f p_t = \Delta r_t,$$

where $\Delta f p_t$ is the change in the forward premium $f p_t = f_t - e_t$ and Δr_t is the change in the interest rate during such window. That is, changes in the forward premium capture changes in rates that are purely due to monetary announcements.⁹

Measuring Hand-to-Mouth Shares

In this second step, I describe the micro data coming from household finance surveys to which I apply Kaplan et al. (2014)'s methodology to measure borrowing constraints, which I also briefly describe. Combining the hand-to-mouth shares coming out from this approach with the cross-country output responses to monetary policy shocks presented above results in next section's empirical moment.

Data. For the bulk of countries I recur to European Central Bank's *Household Finance and Consumption Survey (HFCS)*, which collects data at the household level on sociodemographics, consumption and financial positions. Its 2020-2022 wave was conducted in 22 countries, all of which are in our sample: Austria, Belgium, Croatia, Cyprus, Czechia, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Portugal, Slovakia, Slovenia and Spain. For non-European countries, I recur to their respective household finances surveys: China's *China Household Finance Survey (CHFS)*'s 2011 wave, Mexico's *Encuesta Nacional sobre las Finanzas de los Hogares (ENFIH)*'s 2019 wave, Russia's *All-Russian Survey of Consumer Finances (RUS)*'s 2018 wave, South Africa's *National Income Dynamics Study (NIDS)*'s 2014/2015 wave and Thailand (*THH*)'s *Household Annual Resurvey*'s 2012 wave. Most of these surveys closely follow United States' *Survey of Consumer Finances* design.

Methodology. Questions are not exactly the same across surveys, but are similar enough to create "buckets" of assets and liabilities, which can themselves be grouped into liquid and illiquid wealth. Following Kaplan et al. (2014), the former includes cash, stocks, bonds, checking and saving accounts, net of credit card debts. The latter includes real estate and pension plans net of mortgages and insurance plans. Kaplan et al. (2014) classify household i at time t as poor hand-to-mouth if

⁹ Since the Great Recession, a gap has opened in the CIP (see Du and Schreger 2022). Appendix D shows that the instrument is robust to such deviations since they remain constant during the narrow window.

$$b_{it} \leq 0, \quad \text{and} \quad 0 \leq a_{it} \leq \underline{a}$$

and as wealthy hand-to-mouth if

$$b_{it} > 0, \quad \text{and} \quad 0 \leq a_{it} \leq \underline{a}$$

where b_{it} is net illiquid wealth, a_{it} is net liquid wealth and $\underline{a}$ is the borrowing constraint, that they set to be half monthly labor income.¹⁰ Therefore, a household is classified as hand-to-mouth if it was classified as poor hand-to-mouth or wealthy hand-to-mouth. That is, a household is hand-to-mouth when it does not have enough net liquid wealth to offer as collateral for a credit that would allow for consumption smoothing. The key difference with previous measures that did not distinguish between liquidity types is that now households with positive illiquid wealth $b_{it} > 0$ are considered hand-to-mouth too, ramping up the number of constrained households. Previous hand-to-mouth shares measures were lower bounds since net wealth included illiquid assets $0 \leq a_{it} + b_{it} \leq \underline{a}$. For all purposes, wealthy hand-to-mouth households behave as poor hand-to-mouth. Ignoring them would understate the prevalence of borrowing constraints, leaving variation in the distribution of MPCs unexplained.

2.2 Macro Evidence: Results

The methodologies outlined above are one possible measure of aggregate shocks responses and development at the country level. Before showing the estimates coming from these methods, I first provide validation for them.

Validation

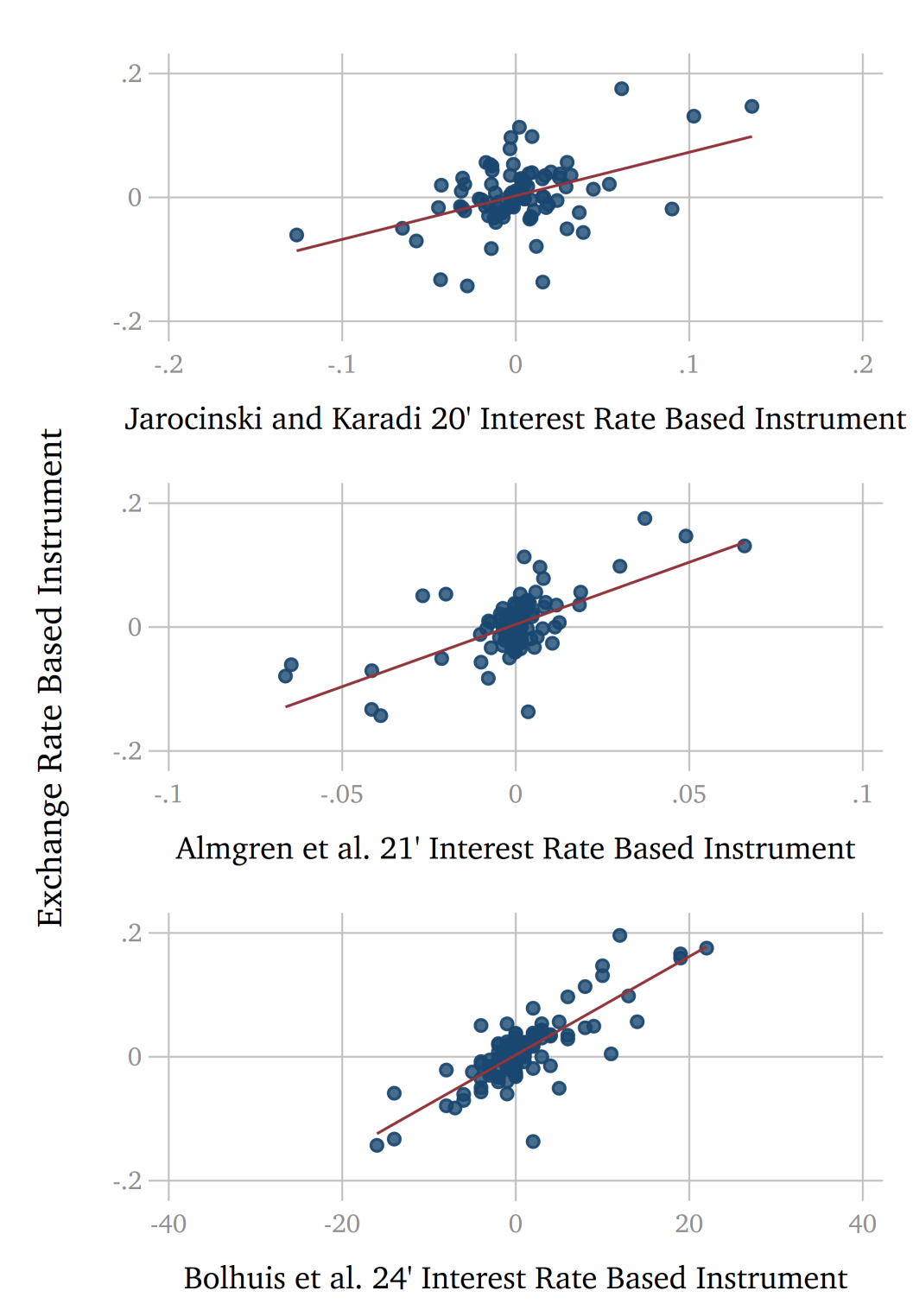
In this section, checks for the empirical strategy are carried out before delving into its results.

External Instrument. A virtue of the exchange rate futures based instrument is that one can construct shocks for countries that do not have interest rates futures. In order to validate this proxy, I take advantage of the fact that both types of markets exist for the Euro. Figure 1 shows that the instrument constructed from Euro forward and spot exchange rates correlates strongly with instruments constructed from Euro interest rates futures.

Relevance and exogeneity are sufficient for identification. However, they are not enough for inference. The correlation between the proxy and the variable to be instrumented not only needs to be different from zero, but also strong. Table 1 shows the results from such first stage for all the countries in the sample, for both the interest rate and exchange rate based instruments. Montiel Olea et al. (2021) recommend the F-test to be above 10 as a rule of thumb for Proxy SVAR estimates. This is the case for at least one instrument for each country, and whenever both are available the stronger one is used.

¹⁰ The “half” comes from resources being consumed at a constant rate in Kaplan et al. (2014)’s model.

Figure 1: Euro Exchange Rate Based and Interest Rate Based Instruments



Notes. Upper panel shows a Pearson correlation of 0.44 between Jarociński and Karadi (2020)'s changes in European Central Bank (ECB) interest rates futures around monetary policy announcements and the changes in Euro exchange rates futures around ECB announcements. Middle panel shows a Pearson correlation of 0.66 with Almgren et al. (2022) interest rate based instrument. Lower panel shows a Pearson correlation of 0.81 with Bolhuis et al. (2024) instrument.

Table 1: First Stage Regressions of Interest Rates on External Instruments

Country	Instrument	F-stat	Coefficient	p-value	R^2	Adj. R^2
Austria	Interest Rate	24.593	0.004	0.000	0.176	0.169
	Exchange Rate	14.835	0.267	0.000	0.114	0.107
Belgium	Interest Rate	19.776	0.003	0.000	0.147	0.139
	Exchange Rate	14.383	0.263	0.000	0.111	0.103
China	Interest Rate	20.039	0.003	0.000	0.148	0.141
	Exchange Rate			—		
Croatia	Interest Rate	26.428	0.004	0.000	0.187	0.180
	Exchange Rate	16.467	0.293	0.000	0.125	0.118
Cyprus	Interest Rate	25.636	0.004	0.000	0.182	0.175
	Exchange Rate	16.003	0.291	0.000	0.122	0.115
Czechia	Interest Rate	15.144	2.974	0.000	0.116	0.109
	Exchange Rate			—		
Estonia	Interest Rate	29.007	0.004	0.000	0.201	0.195
	Exchange Rate	20.957	0.327	0.000	0.154	0.147
Finland	Interest Rate	25.041	0.004	0.000	0.179	0.172
	Exchange Rate	16.264	0.284	0.000	0.124	0.116
France	Interest Rate	22.155	0.004	0.000	0.162	0.154
	Exchange Rate	13.894	0.266	0.000	0.108	0.100
Germany	Interest Rate	30.313	0.004	0.000	0.209	0.202
	Exchange Rate	22.027	0.341	0.000	0.161	0.153
Greece	Interest Rate	26.118	0.004	0.000	0.185	0.178
	Exchange Rate	16.387	0.295	0.000	0.125	0.117
Hungary	Interest Rate	28.156	3.481	0.000	0.197	0.190
	Exchange Rate			—		
Ireland	Interest Rate	27.337	0.004	0.000	0.192	0.185
	Exchange Rate	18.624	0.297	0.000	0.139	0.132
Italy	Interest Rate	28.085	0.004	0.000	0.196	0.189
	Exchange Rate	17.666	0.300	0.000	0.133	0.126
Latvia	Interest Rate	29.344	0.004	0.000	0.203	0.196
	Exchange Rate	21.819	0.331	0.000	0.160	0.152
Lithuania	Interest Rate	26.804	0.004	0.000	0.189	0.182
	Exchange Rate	14.449	0.265	0.000	0.112	0.104
Luxembourg	Interest Rate	27.471	0.004	0.000	0.193	0.186
	Exchange Rate	19.885	0.314	0.000	0.147	0.140
Malta	Interest Rate	29.227	0.004	0.000	0.203	0.196
	Exchange Rate	22.150	0.326	0.000	0.162	0.154
Mexico	Interest Rate	18.045	0.005	0.000	0.136	0.128
	Exchange Rate			—		
Netherlands	Interest Rate	30.097	0.004	0.000	0.207	0.201
	Exchange Rate	19.642	0.313	0.000	0.146	0.139
Portugal	Interest Rate	26.620	0.004	0.000	0.188	0.181
	Exchange Rate	16.984	0.294	0.000	0.129	0.121
Russia	Interest Rate	97.505	0.015	0.000	0.499	0.494
	Exchange Rate	54.579	-1.011	0.000	0.358	0.351
Slovakia	Interest Rate	24.255	0.004	0.000	0.174	0.167
	Exchange Rate	18.713	0.289	0.000	0.140	0.133
Slovenia	Interest Rate	34.058	0.004	0.000	0.229	0.222
	Exchange Rate	23.570	0.341	0.000	0.170	0.163
South Africa	Interest Rate			—		
	Exchange Rate	29.430	-1.121	0.000	0.204	0.197
Spain	Interest Rate	33.244	0.005	0.000	0.224	0.218
	Exchange Rate	20.558	0.329	0.000	0.152	0.144
Thailand	Interest Rate	48.270	0.013	0.000	0.296	0.290
	Exchange Rate			—		

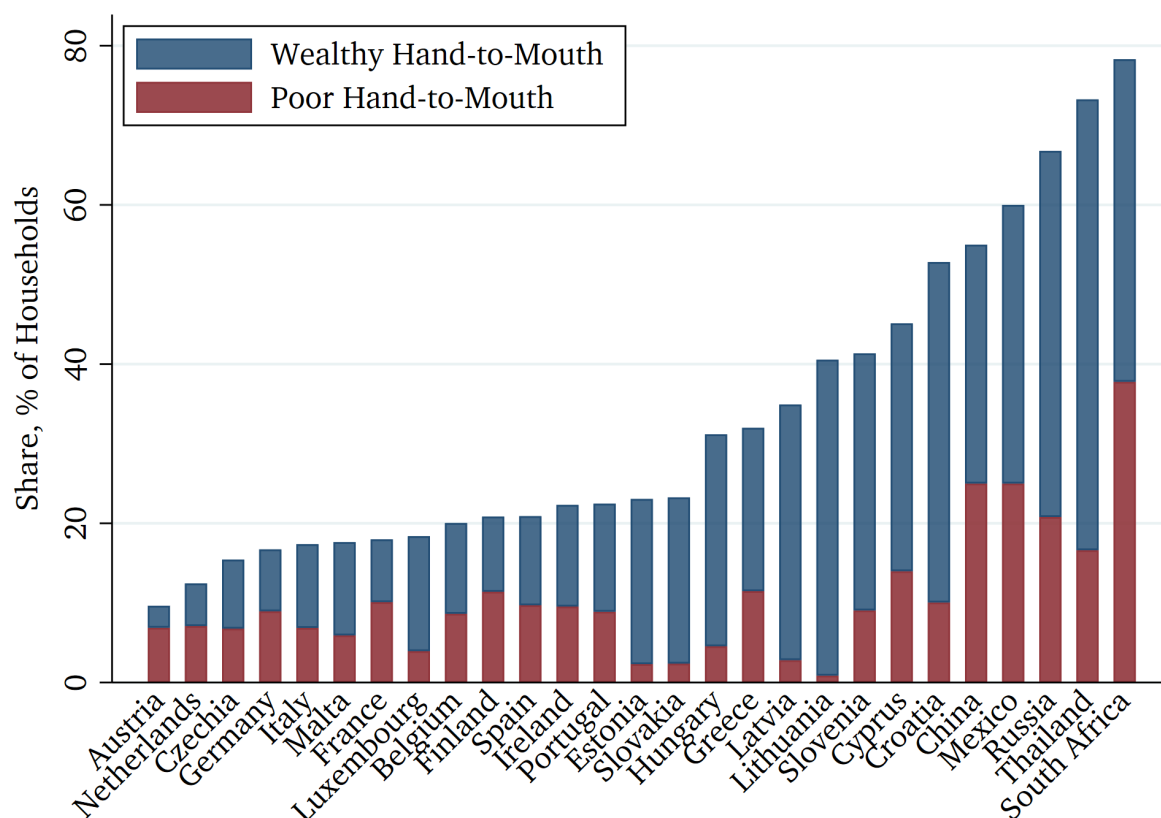
Notes. Results of country-specific first stage regressions between the variable to be instrumented (interest rate) and the instrument (interest rate or exchange rate based).

Baseline Estimates

Having validated these methodologies, I proceed to show the relationship between aggregates responses to shocks and development. Towards that end, I display before the cross-country hand-to-mouth shares calculations and the Proxy SVAR estimated output reactions to monetary shocks.

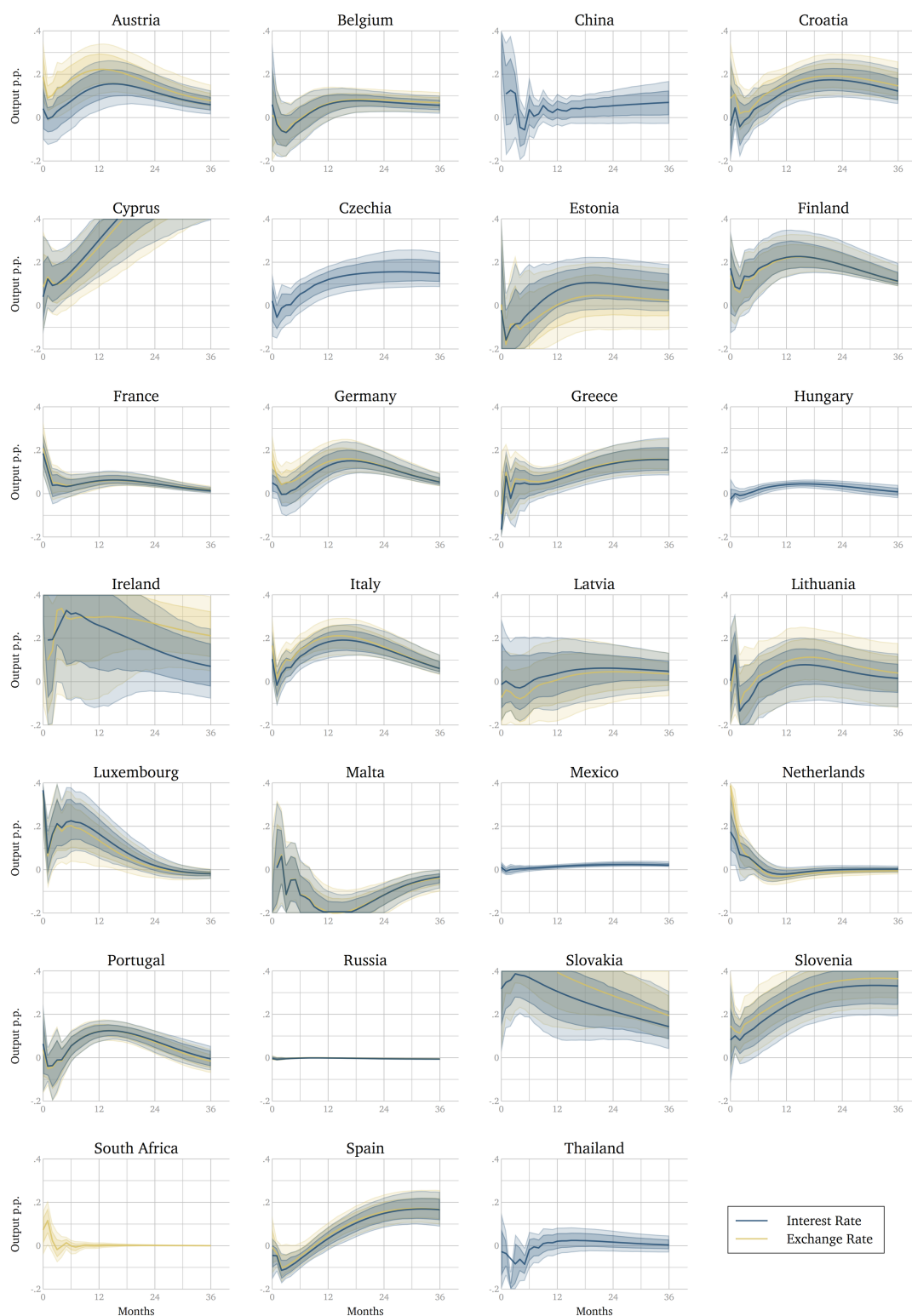
Hand-to-Mouth Shares. Figure 2 presents the poor and wealthy hand-to-mouth shares estimates. The first fact that stands out from the picture is the large cross-country variation in hand-to-mouth shares. They go from around 15% in Italy to nearly 75% in Thailand, and extend even further at the extremes. This covers a wider range than Kaplan et al. (2014), whose minimum and maximum shares were 20% and 30%, and Almgren et al. (2022), with 15% and 62%. Another fact that comes out from the figure is that these estimates are a strong proxy of development: a 10 p.p. increase in the hand-to-mouth share lowers per capita GDP in 7,723 dollars, a correlation significant at the 1% level. A third key fact that emerges from these calculations is the importance of taking into account wealthy hand-to-mouth households. If hand-to-mouth status was only granted to poor hand-to-mouth households, both the level and variation of borrowing constraints would be lower, ranging from 0.9% to 37.8%. Wealthy hand-to-mouth households contribute in both dimensions, ranging from 2.8% to 56.6%.

Figure 2: Hand-to-Mouth Shares



Notes. A household is classified as poor hand-to-mouth if its net liquid wealth is larger or equal than zero but smaller or equal than one-half their monthly labor income and their net illiquid wealth is less or equal than zero. A household is classified as wealthy hand-to-mouth if their net illiquid wealth is larger than zero instead. The sum of the two shares makes for the share of households classified as hand-to-mouth.

Figure 3: Proxy SVAR Output Impulse Response Functions (IRFs) to Monetary Shocks

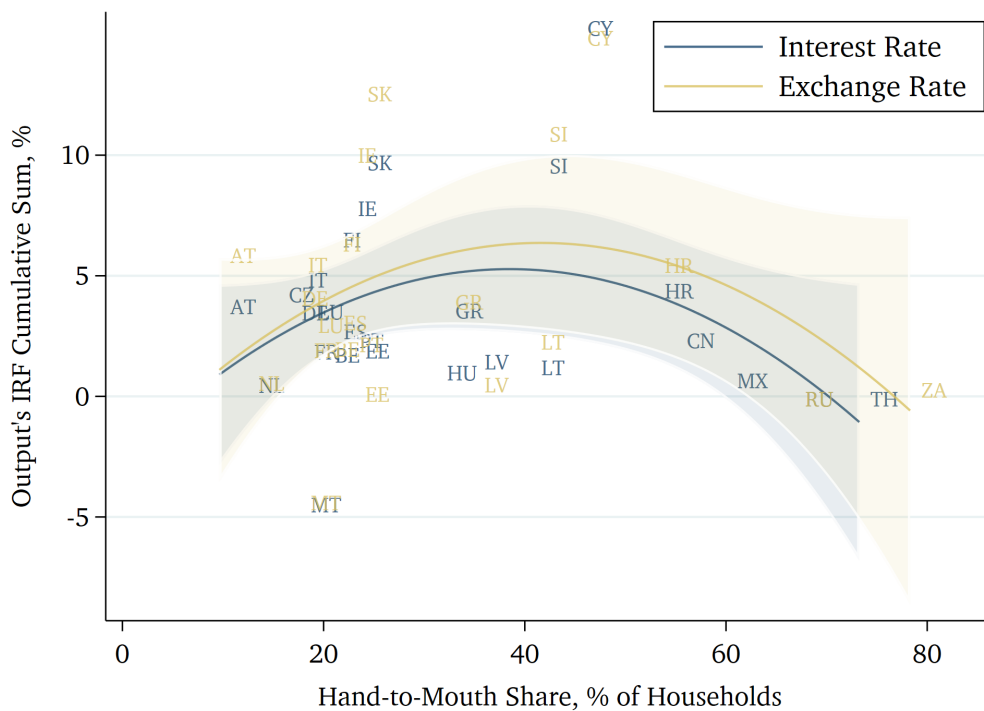


Notes. Lines are impulse response functions of industrial production in logs to a -1.5 p.p. interest rate shock estimated via Proxy SVAR. Unemployment rate and CPI in logs also included in the VAR. Shaded areas are 68% and 90% confidence bands from Jentsch and Lunsford (2019)'s moving block bootstrap standard errors.

Output Responses to Monetary Shocks. Figure 3 plots impulse response functions to a -1.5 p.p. interest rate shock under the baseline Proxy SVAR method, overlaying estimates from both instrument types. Responses are highly similar across proxies, what provides extra validation for the exchange rate based instruments and suggests that they are also purged from any remaining predictable component. A pattern emerges when we consider our three running examples. Italy, which has one of the lowest hand-to-mouth shares, displays a standard hump-shaped positive response. Cyprus, with an average hand-to-mouth share, exhibits large and persistent effects that extend beyond the 36-month window. Thailand, with the second highest hand-to-mouth share, has, if anything, a negative response.

A clearer picture becomes apparent when calculating the cumulative sum of those IRFs and plot them against the hand-to-mouth shares. The result is the inverted U-shaped in Figure 4. The three running examples illustrate the pattern. Italy, with nearly 15% of hand-to-mouth households, has a 5% cumulative sum. Cyprus, which has roughly an equal share of both types, displays nearly three times that effect. Thailand, that mostly has hand-to-mouth households, exhibits a somewhat negative response. The figure overlays estimates coming from both instrument families showing that the moment is robust to the identification strategy. Moreover, in both Figures 3 and 4 the interest and exchange rate instruments estimates closely track each other, suggesting that the predictable component that was purged from the former is neither present in the latter.

Figure 4: Proxy SVAR Output Cumulative IRFs and Hand-to-Mouth Shares



Notes. Lines are OLS estimated quadratic relationships between the cumulative sum of impulse response functions of industrial production in logs to monetary shocks estimated via Proxy SVAR and hand-to-mouth shares. A household is classified as such if its net liquid wealth is larger or equal than zero but smaller or equal than one-half their monthly labor income. Shaded areas are 95% confidence bands.

More formally, Table 2 shows regressions of the form

$$\hat{C}^c = \alpha + \beta_1 \text{HtM}_c + \beta_2 \text{HtM}_c^2 + \mathbf{X}_c' \boldsymbol{\gamma} + \varepsilon_c$$

where $\hat{C}^c = \sum_{h=0}^T \hat{\theta}_h^c$ is the cumulative sum of the estimated impulse response functions for country c , HtM_c is the hand-to-mouth share, and $\mathbf{X}_c$ is a vector of controls from *World Bank Structural Indicators (WBSI)* including per capita GDP, share of exports and share of savings. Although even the plain unconditional correlation measured in column 1 of Table 2 might be of interest, things other than borrowing constraints could differ across countries. Column 2 then adds the *WBSI* controls. The quadratic relationship becomes stronger and R^2 increases. Column 3 drops the two highest hand-to-mouth share countries, Russia and Thailand. Coefficients become imprecise, but larger. Column 4 uses the exchange rate instrument instead of the interest rate one. The magnitudes are sizable, albeit not statistically significant at the 5%.

The cumulative IRFs on the left-hand side of these regressions, $\hat{C}^c$, are estimated objects. Since the standard errors in Table 2 do not account for the estimation uncertainty in the IRFs themselves, Appendix F presents a simulation exercise that does. For each bootstrap iteration, a random cumulative response $\tilde{C}^{c,b}$ is drawn uniformly from within the 90% Proxy SVAR confidence band for every country, and the cross-country quadratic regression is re-estimated with these draws as the dependent variable. Across 10,000 iterations, the hump shape ($\hat{\beta}_1 > 0$, $\hat{\beta}_2 < 0$) is preserved in over 99.9% of draws for both instrument types, and the median bootstrap coefficients closely track the point estimates. Beyond being a robustness check, this exercise suggests that Table 2's fragile statistical significance responds more to the lack of statistical power than to the magnitudes of the coefficients not being sizable.

Table 2: Proxy SVAR Cumulative IRFs and Hand-to-Mouth Shares

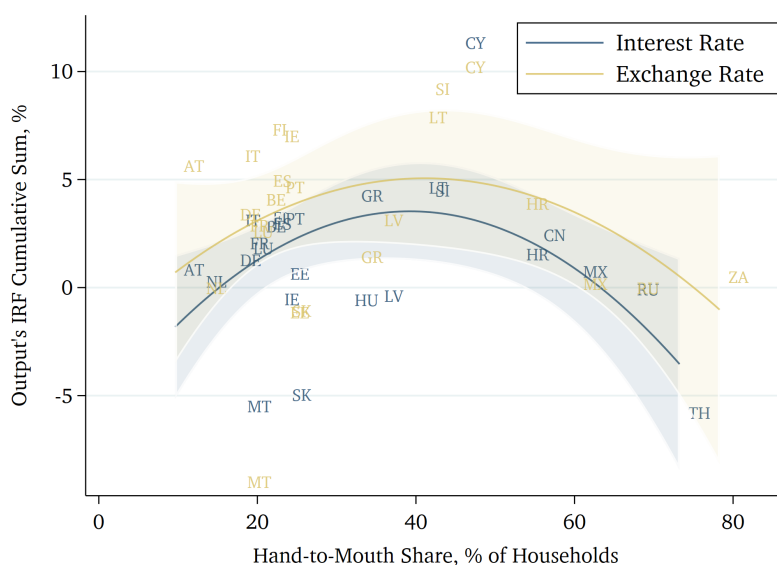
	(1)	(2)	(3)	(4)
	IRF Cum. Sum	IRF Cum. Sum	IRF Cum. Sum	IRF Cum. Sum
HtM Share	40.444* (22.956)	48.559* (25.189)	54.463 (34.275)	52.538 (30.819)
HtM Share Squared	-52.523* (26.302)	-56.517** (26.978)	-62.579 (43.529)	-59.711* (30.692)
Observations	26	26	24	22
R^2	0.150	0.239	0.217	0.195
Controls	No	Yes	Yes	Yes
Instrument	Interest Rate	Interest Rate	Interest Rate	Exchange Rate

Notes. Results of OLS estimated cross-country quadratic relationships between aggregates reaction to shocks and development. The former is given by the cumulative sum of impulse response functions of industrial production in logs to monetary shocks estimated via Proxy SVAR. The latter is measured as hand-to-mouth shares, where a household is classified as such if its net liquid wealth is larger or equal than zero but smaller or equal than one-half their monthly labor income. Controls from *WBSI* include per capita GDP, share of exports and share of savings. All regressions include a constant. Robust standard errors in parentheses.

2.3 Macro Evidence: Alternative Methods

In spite of being widely used, Proxy SVAR results could be sensitive to model specification (see Coibion 2012, Ramey 2016). VAR methods get the benefit of considerably reducing variance by imposing a particular structure. However, this comes at the cost of potentially introducing bias (see Stock and Watson 2018). To address these concerns, Local Projections identification and Bayesian estimation methods estimates are presented. Figure 5 replicates Figure 4 but under Local Projections IV. Table 3 reestimates column 2 of Table 2, but with the cumulative sum of Local Projections IV, Proxy SVAR Local Projections, Bayesian Local Projections and Bayesian Proxy SVAR IRFs as outcomes. Results are broadly similar to the ones obtained with the baseline Proxy SVAR method.¹¹

Figure 5: LPIV Output Cumulative IRFs and Hand-to-Mouth Shares



Notes. Lines are OLS estimated quadratic relationships between the cumulative sum of impulse response functions of industrial production in logs to monetary shocks estimated via Local Projections IV and hand-to-mouth shares. A household is classified as such if its net liquid wealth is larger or equal than zero but smaller or equal than one-half their monthly labor income. Shaded areas are 95% confidence bands.

Table 3: Alternative Methods Cumulative IRFs and Hand-to-Mouth Shares

Method	HtM Share [95% CI]	HtM Share Sq. [95% CI]
Proxy SVAR	48.559 [-0.811; 97.929]	-56.517 [-109.394; -3.640]
Local Projections IV	61.057 [-17.437; 139.551]	-60.755 [-141.666; 20.156]
Proxy SVAR Local Projections	35.378 [3.084; 67.671]	-41.561 [-75.715; -7.408]
Bayesian Local Projections	32.204 [-4.069; 68.477]	-34.767 [-70.737; 1.203]
Bayesian Proxy SVAR	47.324 [0.575; 94.073]	-51.303 [-98.056; -4.550]

Notes. Results of OLS estimated cross-country quadratic relationships between aggregates reaction to shocks and development. The former is given by the cumulative sum of impulse response functions of industrial production in logs to monetary shocks estimated via Proxy SVAR, LPIV, Proxy SVAR LP, Bayesian LP and Bayesian Proxy SVAR. The latter is measured as hand-to-mouth shares, where a household is classified as such if its net liquid wealth is larger or equal than zero but smaller or equal than one-half their monthly labor income. Controls from WBSI include per capita GDP, share of exports and share of savings. All regressions include a constant. 95% CI from robust standard errors.

¹¹ Further discussion of these methods can be found in Appendix E.

2.4 Micro Evidence: Household Consumption Responses

As a segue to the theoretical model, I exploit the panel dimension of the micro data to show heterogeneity in households' consumption responses. Along the way, I also estimate average responses, which point to an inverted U-shaped relationship at the individual level. This subsection follows Holm et al. (2021).

Data. Unfortunately, panel data is not available for all 27 countries in the sample. Fortunately, one country at each stage of development has a panel structure: Italy's *Survey of Household Income and Wealth (SHIW)*'s 2010-2022 biennially, Cyprus' *HFCS*'s 2014-2020 triennially and Thailand's *Household Annual Resurvey (THH)*'s 2009-2015 annually. Interest rates are listed in Appendix A and are the ones that were used for the macro estimates.

Methodology. I separately estimate for each country LPIV regressions of the form

$$\log c_{i,t+h} = \alpha_i + \beta_h \hat{r}_t + \delta \log c_{i,t-1} + \varepsilon_{i,t+h}, \quad h = 0, 1, \dots, H, \quad (1)$$

where $c_{i,t}$ denotes household i consumption at period $t + h$, α_i is a household fixed effect and $\varepsilon_{i,t+h}$ is an error term. The coefficient β_h captures the average consumption response at horizon h from a -1 percentage point change at period t in the predicted interest rate $\hat{r}_t$, which comes from the country specific first stage regressions

$$r_t = \alpha_i + \pi z_t + \phi \log c_{i,t-1} + v_{i,t},$$

where r_t is the interest rate at period t , z_t is the external instrument and $v_{i,t}$ is an error term.

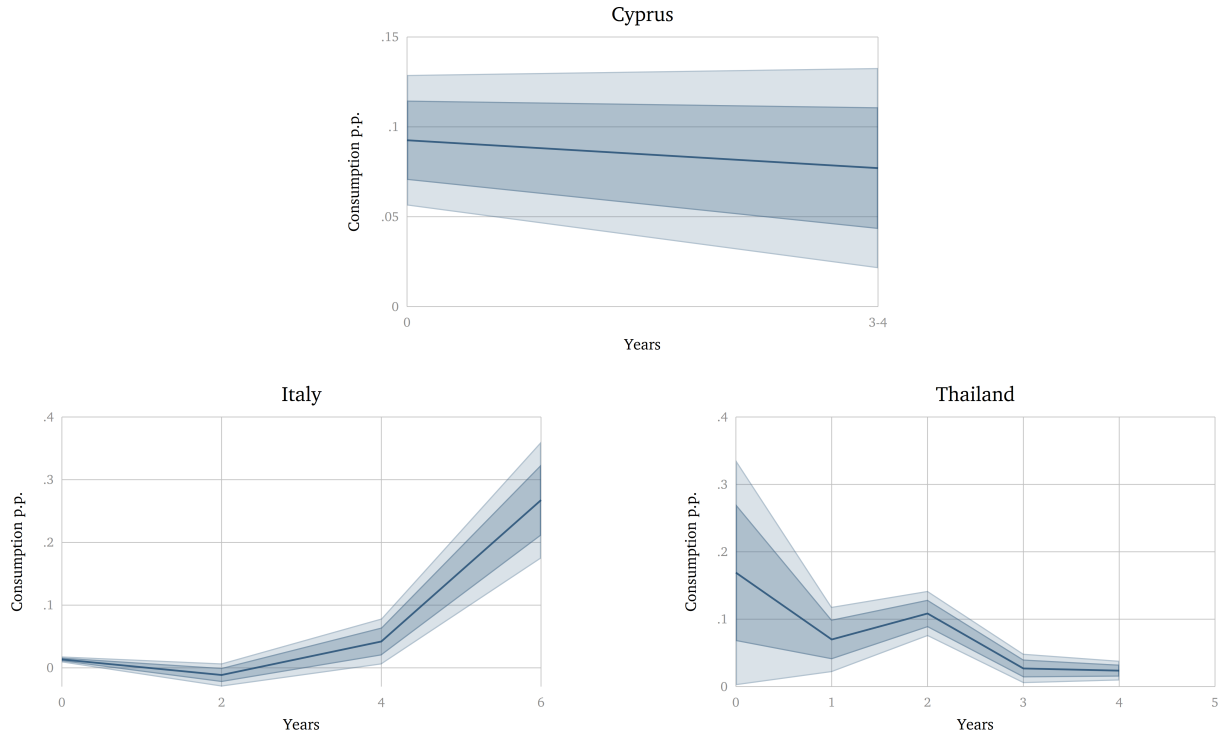
External Instrument. I exploit the interest rate based instrument since the exchange rate one is not available for Thailand. Table 1 already validated this instrument for the three countries in the macro setting. In this micro setting, Kleibergen-Paap F-statistics exceed 10,000 for all three countries, reflecting the large number of observations available.

Average Responses. Figure 6 displays the average household consumption response β_h from equation 1 for Italy, Cyprus and Thailand. Magnitudes differ across countries, but the message that emerges is that there are statistically significant *micro* responses to monetary shocks. Magnitudes also differ across time. Although one should be cautious when interpreting the timing of these responses given the varying survey frequencies; Italy's response takes some time to build, Cyprus' is sizable and persistent, and Thailand's is modest and erratical (note that the large $h = 0$ response is not statistically significant).

Unfortunately, the varying survey frequency also prevents us from computing the cumulative sum of the impulse response functions and comparing them across countries. Fortunately, the three countries' horizons coincide at four years: $h = 2$ for Italy, $h = 1$ for Cyprus and $h = 4$ for Thailand. Given the specification, taking the response at only one horizon has a natural interpretation. The cumulative sum captured how much *total* consumption a household gains or loses over the entire adjustment path. The coefficient at a single horizon instead tells us where consumption *stands* at that point in time relative to where it would have been absent the shock. Figure 7 plots the average household consumption response at the four-year horizon against the hand-to-mouth share. The micro evidence also suggests an inverted U-shape between economic shocks and development.¹²

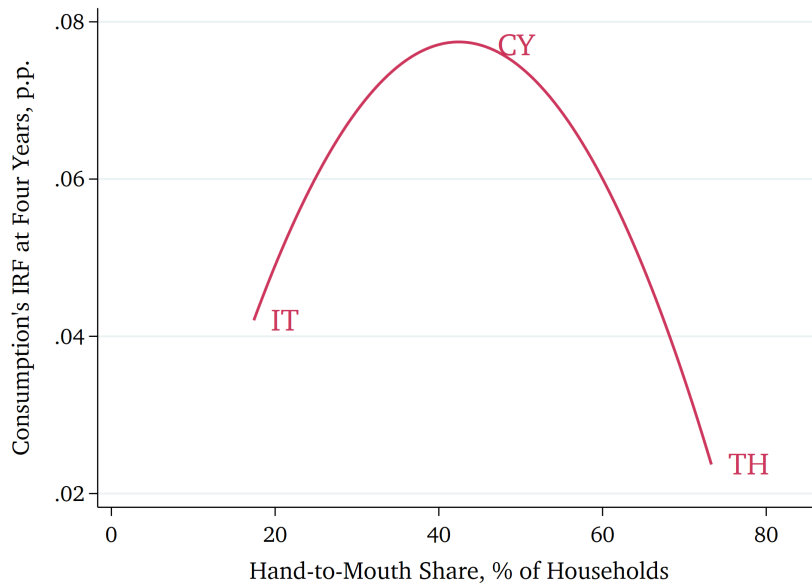
¹² Figure 4 replicates at different horizons rather than as cumulative sums. See Appendix G.

Figure 6: Average Household Consumption IRFs to Monetary Shocks



Notes. Lines are impulse response functions of consumption in logs to a 1 p.p. interest rate decrease estimated via LP-IV with the Bolhuis et al. (2024) high-frequency monetary shock as instrument. Household fixed effects and lagged log consumption (Italy and Thailand) also included. Standard errors are clustered at the household level. Shaded areas are 68% and 90% confidence bands.

Figure 7: Fourth Year Households Consumption IRFs and Hand-to-Mouth Shares



Notes. Line is an OLS estimated quadratic relationship between impulse response functions of consumption in logs at the four-year horizon ($h = 2$ for Italy, $h = 1$ for Cyprus, $h = 4$ for Thailand) to monetary shocks estimated via LP-IV and hand-to-mouth shares. A household is classified as such if its net liquid wealth is larger or equal than zero but smaller or equal than one-half their monthly labor income. Country codes show $\hat{\beta}_h$ from equation 1.

Heterogeneous Responses. I augment the specification with an interaction

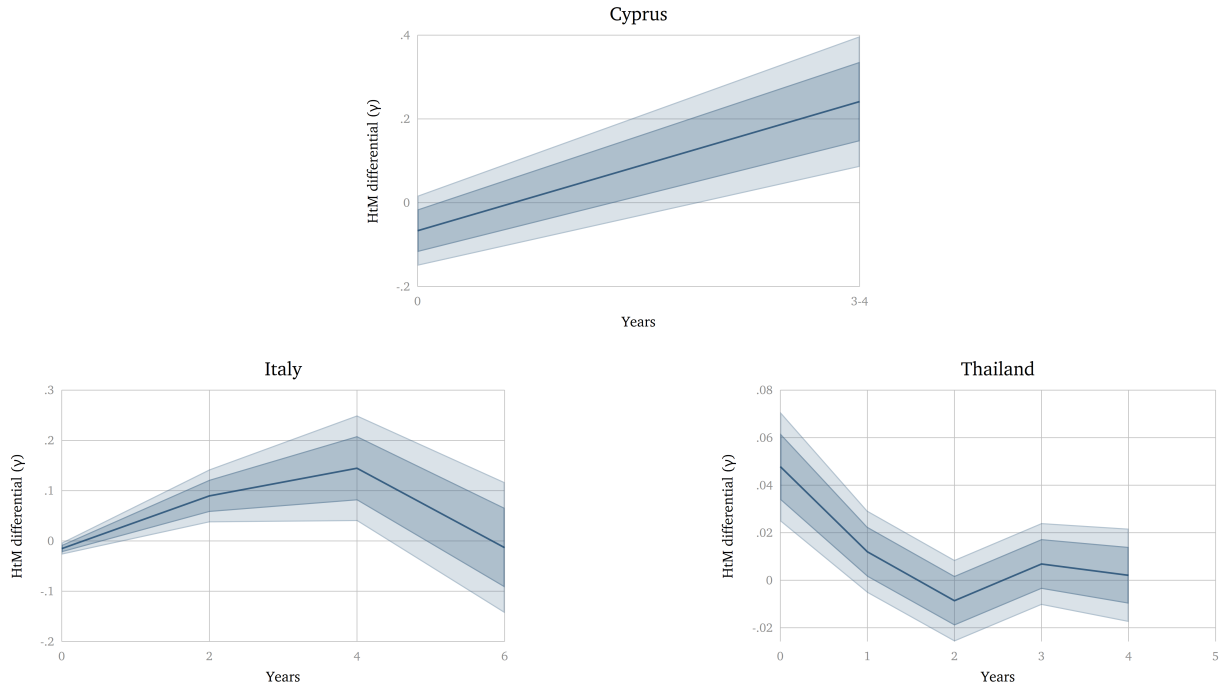
$$\log c_{i,t+h} = \alpha_i + \beta_h \hat{r}_t + \gamma_h \left(r_t \times \widehat{\text{HtM}}_i \right) + \delta \log c_{i,t-1} + \varepsilon_{i,t+h}, \quad (2)$$

where $\text{HtM}_i \in \{0, 1\}$ is the hand-to-mouth status of household i . β_h now captures the response of unconstrained households, while $\beta_h + \gamma_h$ the response of constrained households, with γ_h the differential between the two groups. There are two endogenous variables instrumented in the first stages

$$\begin{aligned} r_t &= \alpha_i + \pi_1 z_t + \pi_2 (z_t \times \text{HtM}_i) + \phi \log c_{i,t-1} + v_{1,i,t} \\ r_t \times \text{HtM}_i &= \alpha_i + \pi_3 z_t + \pi_4 (z_t \times \text{HtM}_i) + \phi \log c_{i,t-1} + v_{2,i,t} \end{aligned}$$

Figure 8 displays the estimated interaction $\hat{\gamma}_h$ from equation 2 at each horizon. In all three countries, hand-to-mouth households exhibit a statistically significant higher response than non-hand-to-mouth households. The timing, all things considered, is suggestive. The Italian interaction response follows a hump-shaped pattern. Its sizable at its peak but dies off. Cyprus has, if anything, the unconstrained households responding more on impact. But the interaction reverses and becomes considerably large in favor of constrained households at the later horizon. In Thailand, the pattern reverses, in the sense that the interaction is positive on impact and fades thereafter. However, note that the magnitudes are small. All of this is consistent with the macro evidence.¹³

Figure 8: Differential Hand-to-Mouth Households Consumption IRFs to Monetary Shocks



Notes. Lines are impulse response functions of the hand-to-mouth differential $\hat{\gamma}_h$ from equation 2 to a 1 p.p. interest rate decrease estimated via LP-IV. A positive γ_h implies hand-to-mouth households respond more strongly than non-hand-to-mouth households at horizon h . Household fixed effects and lagged log consumption (Italy and Thailand) also included. Standard errors are clustered at the household level. Shaded areas are 68% and 90% confidence bands.

¹³Following Holm et al. (2021), Appendix H controls for income to separate direct from indirect effects. The interaction gets attenuated, suggesting that the response is driven by general equilibrium effects.

3 Theory

An inverted U-shaped relationship between output responses to monetary shocks and hand-to-mouth shares was the main result of the empirical section. In this theoretical section, I use that moment to discriminate between models and find that it can only be matched by a one-asset HANK model featuring countercyclical inequality and income risk. The intuition is that countercyclical inequality is an amplification mechanism that complements with the incomplete markets structure in developed countries, but they substitute each other in developing countries. I first lay down the one-asset HANK model with countercyclical inequality. Then, I show that the inverted U-shape can be generated by changing the parameter that controls borrowing constraints. Finally, I demonstrate that other features like fiscal policy or investment cannot by themselves deliver such result.

3.1 Baseline Model

In this sub-section I establish a minimalistic combination between a heterogeneous agent model in the tradition of Bewley (1986), Aiyagari (1994) and, specially, Huggett (1993) with a textbook new keynesian model. Then I explain the calibration strategy and how the steady-state and transitional dynamics are computed.

One-Asset HANK Model

Environment and Behavior. The economy is populated by a unit continuum of infinitely lived ex-ante identical households, who face idiosyncratic but not aggregate risk

$$\begin{aligned} \max_{c_{it}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it}) \\ c_{it} + a_{it} &\leq (1 + r_t) a_{it-1} + e_{it} n_{it} \\ a_{it} &\geq \underline{a}, \end{aligned}$$

where c_{it} is consumption of household i at period t . Households are endowed with productivities e_{it} that follow a Markov process with transition matrix Π . To insure against such idiosyncratic risk, households can borrow against each other up to $\underline{a}$ in an asset a_{it} that pays interest r_t .¹⁴ Income is given by $y_{it} = \frac{W_t}{P_t} e_{it} n_{it}$, but since the production side of this economy is given by a representative firm with a production function

$$Y_t = N_t,$$

real wages are given by $\frac{W_t}{P_t} = 1$ with flexible prices. Nominal wages are sticky instead. This can be microfounded with unions as in Erceg et al. (2000), but I will assume an ad-hoc labor allocation rule as in Auclert and Rognlie (2018)¹⁵

$$n_{it} = Y_t \frac{(e_{it})^{\zeta \log Y_t}}{\mathbb{E} \left[e_i^{1+\zeta \log Y_t} \right]} \equiv Y_t \Gamma(e_{it}, Y_t),$$

¹⁴ The current model abstains from specifying other sources of liquidity like government issued bonds, so liquidity is solely provided by the private agents of this Huggett economy.

¹⁵ Two reasons justify these assumptions. First, because there is no disutility from working, labor needs to be rationed in some way. Second, sticky prices imply that there are profits that need to be distributed, what could implicitly assume some form of cyclical inequality and income risk.

what provides a parsimonious way to control the cyclicity of inequality and income risk.¹⁶ Inequality and income risk will be procyclical when $\zeta > 0$, acyclical when $\zeta = 0$ and countercyclical when $\zeta < 0$. In particular, the standard deviation of household income y_{it} goes up as aggregate income Y_t increases when $\zeta > 0$, the standard deviation of y_{it} remains constant as Y_t increases when $\zeta = 0$ and the standard deviation of y_{it} goes down as Y_t increases when $\zeta < 0$.¹⁷

The path of interest rates follows a real rate rule

$$r_{t+1} = r + \epsilon_t.$$

Which is no other thing than a Taylor rule with coefficient equal to 1

$$\dot{i}_t = r + \pi_{t+1} + \epsilon_t,$$

where r is the steady state real interest rate, ϵ_t is a monetary shock and π_t is price inflation. Since $\frac{W_t}{P_t} = 1$, price inflation is equal to wage inflation π_t^W . The latter is given by

$$\pi_{t+1}^W = 0,$$

an ad-hoc Phillips Curve that closes the model by specifying a supply side.

Equilibrium. Given an exogenous sequence for interest rates r_t , an initial nominal wage W_0 , an initial price level P_0 and an initial distribution of households $\mathcal{D}_0$ over state variables e and a , a competitive equilibrium is given by sequences of prices $\{P_t, W_t, \pi_t, \pi_t^W\}$ and quantities $\{Y_t, C_t, N_t, A_t\}$ and a distribution of households $\mathcal{D}_t(e, a)$ such that households optimize, the representative firm optimizes and goods and assets markets clear

$$\begin{aligned} C_t &\equiv \int c_{it} di = Y_t = N_t = \int n_{it} di \\ A_t &\equiv \int a_{it} di = 0. \end{aligned}$$

Calibration and Computation

I calibrate the model to standard parameters from the literature except for one: the borrowing constraint $\underline{a}$. I will vary this according to the hand-to-mouth shares calculated in empirical section 2. β will be calibrated internally by targeting asset market clearing (with goods market being cleared by Walras' law). The key untargeted moment is the inverted U-shaped relationship between cumulative responses to monetary shocks and hand-to-mouth shares. A positive relationship between MPCs and hand-to-mouth shares is also an untargeted moment. However, this micro fact is of second order importance relative to the aforementioned macro fact, given that it will not allow us to discriminate between different specifications of cyclicity. Table 4 displays the parametrization.

¹⁶ It is well understood (see Acharya and Dogra 2020, Ravn and Sterk 2021 and Bilbiie 2024) that the cyclicity of inequality goes hand in hand with the cyclicity of income risk. The difference between the two is that the former has to do *current* shocks, while the latter with *future* shocks.

¹⁷ This model nests the so-called “canonical” HANK model (see Auclert et al. 2024): under the $\zeta = 0$ acyclical case the labor allocation rule collapses to $n_{it} = N_t$.

Table 4: Calibration of One-Asset HANK Model

Parameter		Value	Reason/Moment	Target
<i>A. Externally calibrated</i>				
σ	IES	0.5	Literature	0.5
ρ_e	Prod. shocks persist.	0.92	Floden and Lindé (2001)	0.9136
σ_e	Prod. shocks std. dev.	0.92	-	-
n_e	Income grid	11	-	-
Y	Aggregate income	1	Normalization	-
ζ	Cyclicalit	-1.1	Validation	-1.1
$\underline{a}$	Borrowing constraint	[-2.99, -1.5, -0.1]	Hand-to-mouth shares	[0.17, 0.41, 0.78]
r	Interest rate	0.01	Annual real interest rate	0.04
<i>B. Internally calibrated</i>				
β	Discount rate	-	Asset market clearing	0

Households have CES utility over consumption $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$. The intertemporal elasticity of substitution (IES) σ is set to 0.5.¹⁸ Floden and Lindé (2001) estimated an autoregressive coefficient for residualized wages of 0.9136. Hence our calibration of ρ_e to 0.92. We also set the standard deviation of such income process to $\sigma_e = 0.92$. With these parameters in hand we discretize the income process into a 11-point Markov chain n_e with Rouwenhorst method. Aggregate income Y is normalized to one.

The only parameters left to be set are the cyclicalit ζ and the borrowing constraint $\underline{a}$. Although I will show results for the three cyclicalit scenarios, the baseline will be the countercyclical case ($\zeta = -1.1$, the empirically relevant according to Guvenen 2009 and the validation below). Borrowing constraints are informed by the average ratio of household's net liquid wealth to annual labor income, from which hand-to-mouth shares were calculated in section 2. In particular, $\underline{a}$ will take three values: -2.99, -1.5 and -0.1. They correspond to the ratio's maximum, average and minimum values, which themselves correspond to Italy (17% hand-to-mouth share), Cyprus (45%) and Thailand (73%). The steady-state interest rate r is set to 0.01.

In order to solve for the steady state of the model and compute transitional dynamics to MIT shocks, I utilize sequence-space Jacobians methods from Auclert et al. (2021a). The steady-state of the model is given by goods market clearing, which can be written as¹⁹

$$Y_t = \mathcal{C}_t(\{r_s, Y_s\})$$

where $\mathcal{C}$ is an aggregate consumption function that maps current and future realizations of r and Y to present consumption. To study transitional dynamics, I feed a shock $\{dr_t\}$

$$dY_t = \sum_{s=0}^{\infty} \frac{\partial \mathcal{C}_t}{\partial r_s} dr_s + \sum_{s=0}^{\infty} \frac{\partial \mathcal{C}_t}{\partial Y_s} dY_s$$

and solve the fixed point problem by computing Jacobians $\mathcal{J}_{t,s}^{\mathcal{C},Y} = \partial \mathcal{C}_t / \partial Y_s$.²⁰

¹⁸ Werning (2015) shows that, with $\sigma = 1$, the response to monetary shocks is the same for any level of liquidity in a HANK than in RANK model. This is so because he further imposes acyclicalit of income risk.

¹⁹ Asset market clears by Walras' law.

²⁰ For details, see Appendix I.

3.2 Results

The baseline model outlined above will answer the question of how aggregates respond to a shock at different stages of development. The answer will depend on the model's calibration, so I validate the parametrization above before delving into results.

Validation

I first try to find some support for the countercyclical calibration in the household finances' dataset. Then I check whether the calibrated model can deliver consistent micro predictions: MPCs that are increasing in borrowing constraints.

Countercyclicality. We saw above how ζ controls the cyclicity of inequality and income risk and we will see below how countercyclical inequality is needed to match section 2's moment. I exploit the micro data structure to empirically back up the calibration $\zeta = -1.1$, since it surveys households in different quarters throughout a year. Then, I can match households income during a particular quarter with aggregate income during that same quarter and answer whether the standard deviation of households income is positively associated with aggregate income (procyclical case), negatively associated (countercyclical case) or not associated (acyclical case).

Formally, given household income $y_{it} = e_{it}n_{it}$ and Auclert and Rognlie (2018) labor allocation rule $n_{it} = Y_t (e_{it})^{\zeta \log Y_t} / \mathbb{E} \left[e_i^{1+\zeta \log Y_t} \right]$ one can show that

$$\text{sd}(\log y_{it}) = (1 + \zeta \log Y_t) \text{sd}(\log e_i),$$

which gives us a reduced-form to test²¹

$$\text{sd}(\log y_{it}) = \text{sd}(\log e_{it}) + \zeta [\text{sd}(\log e_{it}) * \log Y_t] + \varepsilon_t.$$

Taking this equation to the micro data, $\text{sd}(\log y_{it})$ is the standard deviation of log annual labor income in quarter t , $\text{sd}(\log e_{it})$ is the standard deviation of log residualized annual labor income²² and $\log Y_t$ is the log of aggregate income in quarter t .²³ Table 5 displays the results. The negative and statistically significant at the 1% coefficient in the first column can be considered as evidence in favor of countercyclical inequality and income risk. The spirit of the reduced form equation is at the country level, so I bring in country fixed effect to its empirical counterpart in the second column. The result remains statistically significant at the 1% and becomes even larger in magnitude. This is the preferred specification and the coefficient on $[\text{sd}(\log e_{it}) * \log Y_t]$ informs our countercyclical calibration of $\zeta = -1.1$.²⁴

²¹ Note that in the empirical specification below $\text{sd}(\log e_i)$ has an additional t subscript with respect to its theoretical counterpart above. This is so because, in the model, the standard deviation of productivities is fixed through time, while this is not true in the data and we will have to “control” for it.

²² Coming from regressing log annual labor income on household head's age (both linearly and quadratically), gender, education, employment status and number of household members.

²³ Quarterly, seasonally adjusted, real gross domestic product from *IMF IFS*.

²⁴ No information about when the interviews were conducted was available for China, Germany, Hungary, Luxembourg, Mexico, Netherlands, Russia, Spain and Thailand. The number of quarters for the remaining 17 countries ranges from 3 to 8.

Table 5: Standard Deviation of Household's Income and Aggregate Income

	(1)	(2)
	sd (log y_{it})	sd (log y_{it})
sd (log e_{it})	1.503*** (0.159)	2.341*** (0.656)
sd (log e_{it}) * log Y_t	-0.259*** (0.076)	-1.141** (0.544)
Observations	64	64
R^2	0.820	0.941
Country FE	No	Yes

Notes. Results of OLS estimated relationship between the standard deviation of households' income and aggregate income. The former is given by the standard deviation of log annual labor income at quarter t . The latter is measured as the standard deviation of log residualized annual labor income at quarter t and log GDP per capita at quarter t . All regressions include a constant. Robust standard errors in parentheses.

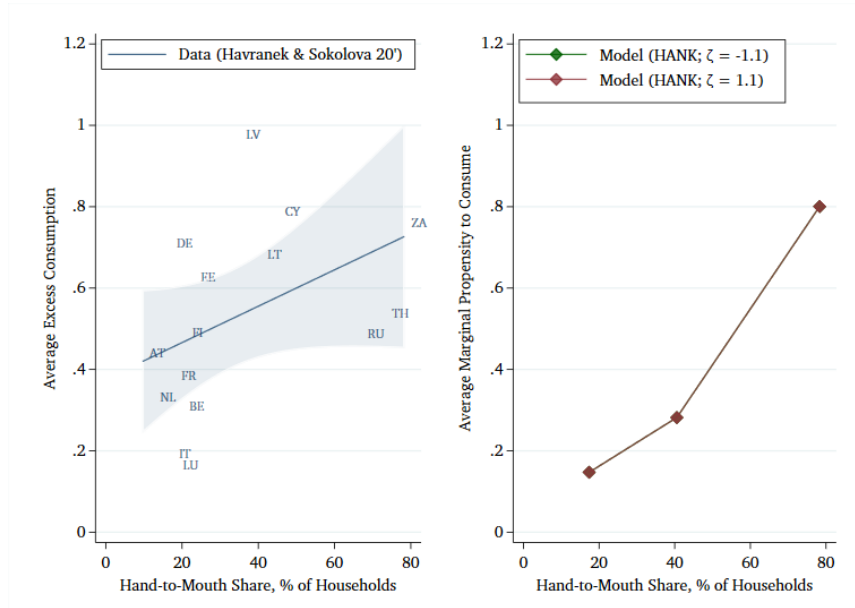
Marginal Propensities to Consume. Micro-level evidence on large propensities to consume out of wealth serves as the keystone empirical fact behind incomplete markets models, since it explicitly contradicts representative agent models' permanent income hypothesis. What this check consists of is comparing the MPCs coming out from the model against the excess consumption estimates compiled in Havranek and Sokolova (2020). They collected 3,127 estimates from the literature of how consumption reacts to anticipated or transitory shocks (which the permanent income hypothesis would predict to be close to zero). For our purposes, we kept only those that come from countries in our sample and averaged across studies (papers usually report more than one estimate) and countries. These average estimates are displayed on the left panel of Figure 9. Meanwhile, the right panel shows average MPCs for the different ζ and $\underline{a}$ calibrations of the model. Both empirically and theoretically there seems to be a positive association between borrowing constraints and propensity to consume.²⁵²⁶

Nevertheless, however useful MPCs are to distinguish between HANK and RANK, it can also be appreciated in Figure 9 that they are not as helpful to discriminate within HANK models: under both cyclical specifications the average MPC is the same. In fact, the result is even stronger: under the two scenarios the distribution of MPCs is exactly alike. This is so because MPCs are steady-state objects and the steady-state is the same under the pro, counter or acyclical cases. Cyclical instead has to do with transitional dynamics, with how the standard deviation of household incomes change when aggregate income changes. Another way to put it is that MPCs answer the partial equilibrium question of how much would a household spend out of windfall income. And the answer to such hypothetical question does not depend on whether the household receives the actual income or not. But, in the countercyclical case for instance, the low-income, high-MPCs households are the ones that in general equilibrium actually receive the extra aggregate demand. Hence, though the model's micro implications are observationally equivalent under any of the three cyclical cases, its macro predictions are not.

²⁵ Excess consumption estimates were not available for China, Croatia, Greece, Hungary, Ireland, Malta, Mexico, Portugal, Slovakia, Slovenia and Spain. The number of available estimates for the remaining 15 countries ranges from 5 to 96.

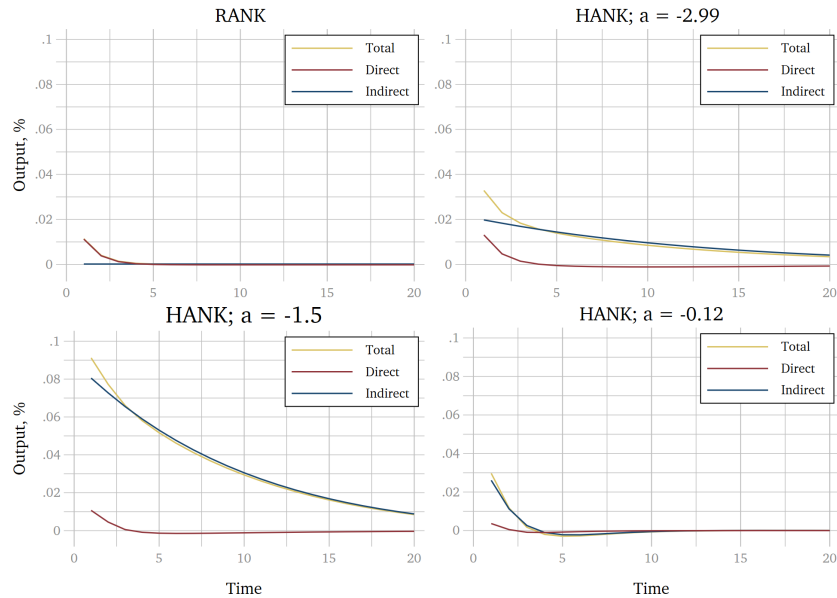
²⁶ One could argue that a quadratic specification would deliver a better fit or that things other than borrowing constraints are different across countries. Appendix J demonstrates that it is not the case.

Figure 9: Marginal Propensities to Consume and Hand-to-Mouth Shares



Notes. Left panel shows a correlation of 0.6 between average excess consumption estimates compiled in Havranek and Sokolova (2020) and hand-to-mouth shares, where a household is classified as such if its net liquid wealth is larger or equal than zero but smaller or equal than one-half their monthly labor income. Right panel shows the relationship between average MPCs and hand-to-mouth shares coming from calibrations of a one-asset HANK model, for different cases of cyclicalities of inequality and income risk ζ .

Figure 10: RANK and One-Asset HANK Output IRFs to Monetary Shock



Notes. Solid lines are impulse response functions of output to a -1.5 p.p. interest rate shock coming from both RANK and HANK models. Dashed and dotted lines respectively decompose them into a substitution effect and an income effect. Northwestern panel shows such results for a RANK model, northeastern panel for a high-liquidity $a = -2.99$ HANK model, southwestern panel for a mid-liquidity $a = -1.5$ HANK model and southeastern panel for a low-liquidity $a = -0.1$ HANK model.

Baseline Results

With the proposed calibration validated, I proceed to show the baseline model results for the relationship between aggregate responses to shocks and development. Figure 10 shows output impulse response functions to an interest rate shock²⁷ under four scenarios: a representative agent model benchmark and the three different $\underline{a}$ parametrizations. The general message that comes out of this figure is that while the heterogeneous agent model delivers somewhat larger responses than the representative agent model under the liquid scenario (a result that is somewhat well known, see Kaplan et al. 2018, Auclert et al. 2024), the models truly differ as we move to the less liquid cases. To understand why, the figure also breaks down the total effects into its direct and indirect effects according to equation (1) above

$$dY_t = \underbrace{\sum_{s=0}^{\infty} \frac{\partial \mathcal{C}_t}{\partial r_s} dr_s}_{\text{Direct Effect}} + \underbrace{\sum_{s=0}^{\infty} \frac{\partial \mathcal{C}_t}{\partial Y_s} dY_s}_{\text{Indirect Effect}}.$$

The first two plots are no other thing than Kaplan et al. 2018's result: while in the RA model the response is completely explained by the direct mechanism, in a (liquid) HANK model the roles are more distributed between direct and indirect effects. The last two plots shed light on this paper's novel result: with intermediate levels of liquidity the direct effects are small and the indirect effects are large, while for low levels of liquidity the direct effects are so small that the indirect effects are not large any more. Another way to put it is that there are still enough agents who intertemporally substitute in the $\underline{a} = -1.5$ case to meaningfully move aggregate demand, kicking down the general equilibrium effect of borrowing constraints being relaxed. For $\underline{a} = -0.1$ the partial equilibrium effect of the few agents on their Euler's equation is not enough to relax the binding borrowing constraints of most of the agents.

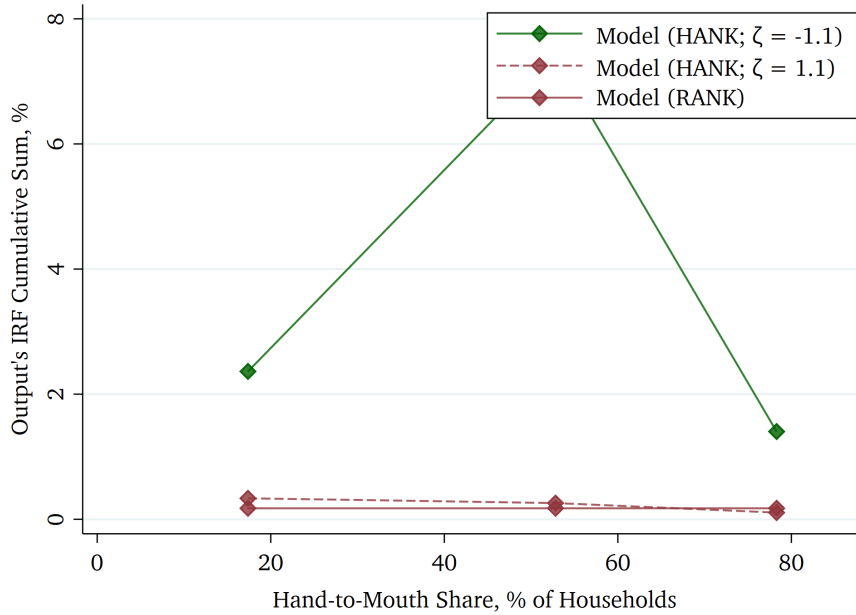
All of this can be seen more clearly in Figure 11. It calculates the cumulative sum of the impulse response functions above for the countercyclical scenario and also brings in the pro and acyclical cases. Note the different cumulative responses' levels within the highest liquidity level. Countercyclicity generates an amplifying income effect by redistributing towards constrained households the additional aggregate demand generated by the substitution effects of unconstrained households. As we move to the lower liquidity levels, this amplification mechanism first complements with the hand-to-mouth share but later they substitute each other since there is less of a direct effect to begin with. Consider instead the $\zeta = 0$ calibration. The potency of monetary policy decreases with borrowing constraints since no indirect effects counteract the direct effects of lower liquidity.²⁸ In the $\zeta = 1.1$ case the direct effects get actually dampened by indirect effects since more households hit their borrowing constraints as aggregate demand increases.

Figure 12 pulls together the empirical and theoretical results. The countercyclical one-asset HANK model matches the inverted U-shaped moment coming from Proxy SVAR estimates. The cross-country relationship between output responses to monetary shocks and hand-to-mouth shares not only favors HANK over RANK, but also disciplines HANK models to feature countercyclical inequality and income risk.

²⁷ Particularly, a -1.5 p.p. interest rate shock with 0.35 persistence.

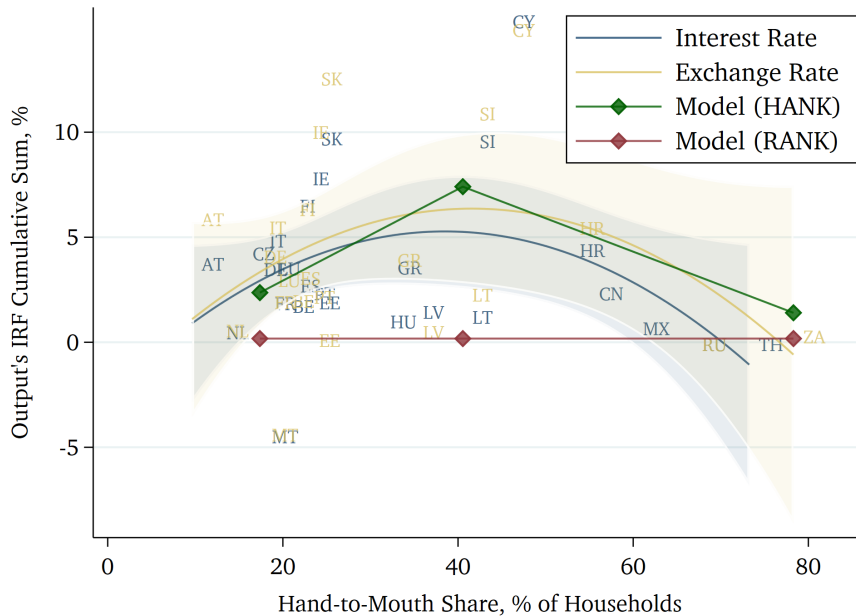
²⁸ Because borrowing constraints are procyclical. If they were acyclical, $\zeta = 0$ would deliver a flat line.

Figure 11: RANK and One-Asset HANK Output Cumulative IRFs to Monetary Shock and Hand-to-Mouth Shares



Notes. Diamonds show the relationship between the cumulative sum of impulse response functions of output to a -1.5 p.p. interest rate shock and hand-to-mouth shares coming from the models.

Figure 12: Proxy SVAR, RANK and One-Asset HANK Output Cumulative IRFs to Monetary Shock and Hand-to-Mouth Shares



Notes. Lines are OLS estimated quadratic relationships between the cumulative sum of impulse response functions of industrial production in logs to monetary shocks estimated via Proxy SVAR and hand-to-mouth shares. A household is classified as such if its net liquid wealth is larger or equal than zero but smaller or equal than one-half their monthly labor income. Shaded areas are 95% confidence bands. Diamonds show the relationship between the cumulative sum of impulse response functions of output to a -1.5 p.p. interest rate shock and hand-to-mouth shares coming from the models.

3.3 Alternative Models

I showed that a one-asset HANK model with countercyclical inequality and income risk can match the inverted U-shape from section 2. In contrast with sub-section 2.3, whose objective was to show how different empirical methods gave way to the same result, the purpose of this sub-section is to show that different models do not deliver the same result. In particular, I will demonstrate that no feature other than countercyclical inequality can replicate the inverted U-shaped moment, unless those features are themselves defined in a countercyclical manner.

One-Asset HANK Model + Fiscal Policy

I augment the one-asset HANK model described above with a government subject to a budget constraint

$$B_t = (1 + r_t) B_{t-1} + G_t - T_t,$$

where B_t are government bonds uniformly bounded to prevent Ponzi schemes, G_t is government spending and $T_t = \tau_t Y_t$ is total tax revenue.²⁹ Total after-tax income is now given by

$$Z_t \equiv Y_t - T_t = (1 - \tau_t) Y_t,$$

while household after-tax income takes the form

$$z_{it} \equiv e_{it} Z_t = e_{it} (1 - \tau_t) Y_t$$

The rest of the model remains the same, with the caveat that we will focus on the acyclical $\zeta = 0$ case. Under such scenario the model boils down to the “canonical” HANK, with a labor allocation rule $n_{it} = N_t$.³⁰ Equilibrium is given by agents optimizing and markets clearing

$$\begin{aligned} C_t + G_t &= Y_t = N_t \\ A_t &= B_t \end{aligned}$$

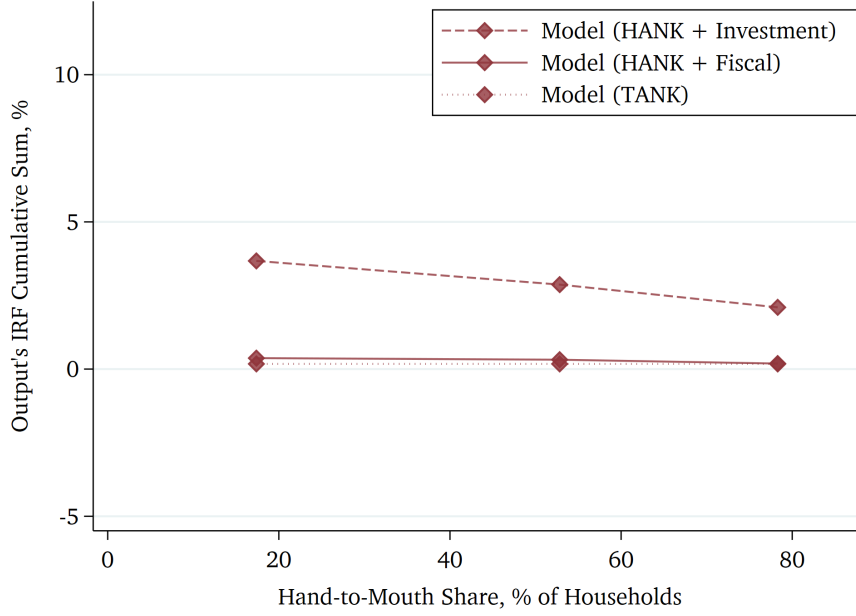
We calibrate $G = 0.2$ and $B = 0.8$ and repeat the exercise above of computing cumulative impulse response functions to a r shock for the different $\underline{a}$ levels.³¹ The results are displayed in Figure 13. We can appreciate that this alternative model generates a negative relationship between the cumulative responses to monetary policy and the tightness of borrowing constraints. This is so because despite ζ being set equal to zero, the linear way in which taxes are introduced makes the standard deviation of household incomes shrink during recessions. That is, inequality and income risk are implicitly specified to be procyclical. How the cyclicalities of inequality and income risk are specified remains therefore the key feature that determines the shape of the relationship between the response to aggregate shocks and development, and there is nothing in particular about fiscal policy that can match the inverted U-shape.

²⁹ Note that these government bonds pay the same interest rate r_t than assets A_t , so for all practical purposes this is still a one-asset model.

³⁰ Note that this assumption is the one that gives you the new household income in contraposition with the old one $y_{it} = e_{it} n_{it}$ since $Y_t = N_t$.

³¹ Note that G is another margin to control liquidity, but I keep it constant as I change $\underline{a}$.

Figure 13: Alternative Models Output Cumulative IRFs to Monetary Shock and Hand-to-Mouth Shares



Notes. Diamonds show the relationship between the cumulative sum of impulse response functions of output to a -1.5 p.p. interest rate shock and hand-to-mouth shares coming from the models.

Two-Agent Model

A simpler model can be taken into consideration. Many of the features from heterogeneous agents models can be captured by two-agent models (see Bilbiie 2020). In these kind of models there is a fraction μ of agents that behaves entirely as hand-to-mouth, while a fraction $1 - \mu$ behaves as a representative consumer

$$c_t^{\text{HTM}} = Y_t$$

$$c_t^{\text{RA}} = \frac{\beta^{t/\sigma} q_t^{-1/\sigma}}{\sum_{s \geq 0} \beta^{s/\sigma} q_s^{1-1/\sigma}} \left[\sum_{s \geq 0} q_s Z_s + (1 + r_0) a_{-1} \right].$$

Aggregate consumption and output are then given by

$$C_t = \mu c_t^{\text{HTM}} + (1 - \mu) c_t^{\text{RA}}.$$

I calibrate $\mu = 0.17$, $\mu = 0.41$ and $\mu = 0.78$ and redo the computations above. Figure 13 shows that the TA model predicts a constant relationship between aggregate responses and development. The intuition for such result is as follows. An expansive monetary shock makes the RA agents intertemporally substitute consumption into this period. This increases aggregate demand Y , what makes HTM agents have more income, which they fully consume. Hence, $c_t^{\text{HTM}} = c_t^{\text{RA}}$, since whatever additional consumption the RA agents make will translate one-to-one to Y and, therefore, to HTM agents consumption. Then, aggregate consumption C_t will be the same independently of μ . Since households never switch from being HTM to RA or viceversa, the model does not feature any cyclicity of inequality and income risk. Again, this feature is crucial for the inverted U-shaped relationship. There is nothing inherent to the TA model that can explain this fact.

Two-Asset HANK Model + Investment

A fully-specified model can also be taken into consideration. Firms now have capital k_{t-1} on their production functions

$$y_{jt} = k_{jt-1}^\alpha n_{jt}^{1-\alpha},$$

which they can choose paying a quadratic adjustment cost that gives way to a Tobin's Q-type law of motion for capital.

A representative intermediary finances that capital by investing in firm equity (and government bonds) the household's deposits. This is a two-asset HANK model since households now have a liquid and an illiquid account as in Kaplan et al. (2018). Illiquid assets can be converted into liquid by paying a fixed cost. The model features price as well as wage stickiness, with optimizing unions microfounding the Phillips curve. Equilibrium still involves agents optimizing and goods and assets markets clearing.

Liquid accounts are a target now and are set according to the previous three $\underline{a}$ levels. Total wealth is a new target, informed by micro data from the same countries that informed liquid accounts. Illiquid wealth is an untargeted moment.³² I redo the computations above and the results can be appreciated in Figure 13 as well. Investment gives way to a strong output reaction for the highest liquidity level. However, the large direct effects hide indirect effects that go in the opposite direction, due to inequality and income risk being procyclical. Moving to the less liquid cases, the direct effects become smaller as the share of hand-to-mouth agents increases. And there are no amplifying indirect effects from countercyclicality to compensate for these lower direct effects. The relationship between output responses to shocks and borrowing constraints ends up being convex instead of concave. Again, there is nothing in particular about investment that can explain the inverted U-shaped relationship.

³² The rest of the model and its parametrization comes directly from Auclert et al. (2021a) and can be found in Appendix K.

4 Conclusions

This paper studies whether the response to aggregate shocks is the same at different stages of development. If that is not the case, business cycles will be different in developed than in developing countries and aggregate demand stabilization policies should take this into account. I find that there is an inverted U-shaped relationship between output reactions to monetary shocks and the share of hand-to-mouth agents. Two empirical challenges needed to be sorted in order to answer such a question: the endogeneity of macroeconomic shocks and developing countries' data scarcity. The former was addressed by exploiting two types of high frequency monetary shocks and the latter by pulling together household finance surveys from high, middle and low-income countries.

Despite the pervasiveness of borrowing constraints in developing countries, incomplete market models do not have definitive predictions on this relationship. I use the empirical moment to dissipate such theoretical ambiguity: a one-asset HANK model with countercyclical inequality and income risk matches it. For any level of development, countercyclicality generates an indirect income effect through redistributing towards low-income, high-MPCs households the additional aggregate demand generated by the direct effect of intertemporal substitution. This amplification is small when there are few constrained agents because general equilibrium effects are small and when there are few unconstrained agents because partial equilibrium effects are small. Amplification peaks instead when there is an equal distribution of constrained and unconstrained agents.

All of the above supports the idea that aggregate responses to shocks might not be the same at different stages of development. For policy shocks, the efficacy of monetary and fiscal aggregate demand stabilization might change when one moves from developed to developing countries. This paper tried to make the case that a complementarity between borrowing constraints and countercyclicality amplifies monetary policy in middle-income countries. Meanwhile, fiscal policy might be dampened in the developing world, since governments cannot finance deficits that would stabilize aggregate demand.³³

For productivity shocks, the trend might be the cycle: structural factors like the prevalence of credit constraints may amplify aggregate demand volatility in response to temporary productivity shocks (above and beyond whatever volatility permanent productivity shocks generate). But the cycle might be the trend too: the inverted U-shape described in this paper may explain why there exists a middle-income trap. When countries achieve average levels of development, they get stuck in a high demand volatility equilibrium from where only a combination of optimal policies and luck can take them to the quieter waters of development.

It should be clear by now that all of this matters because deviations from a representative agent benchmark amplify the welfare costs of business cycles. This paper made the case for borrowing constraints also augmenting the volatility of demand as they become more pervasive. The welfare costs of recessions could be then orders of magnitude higher in non-developed countries.

³³ Sheremirov and Spirovska (2022) estimate fiscal multipliers of 1.6 in developed countries and 0.8 in developing countries. Egger et al. (2022) estimate a multiplier of 2.5 for Kenya because they circumvent the deficit financing conundrum by providing the funds themselves.

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Appendix A

The main exception to *IMF IFS* as a data source concerns interest rates. Table 6 reports the interest rate series used in the VAR for each country. The 12-month or 1-year tenors were chosen to match the 1-year tenor of the overnight indexed swaps underlying the interest rate instruments and the 12-month tenor of the forward contracts underlying the exchange rate instruments.

Table 6: VAR Interest Rates

Country	Interest Rate	Tenor	Source
Euro Area	Euribor	12-month	<i>EMMI</i>
China	Lending Rate	–	<i>Bloomberg</i>
Czechia	<i>PRIBOR</i>	1-year	Czech National Bank
Hungary	<i>BUBOR</i>	1-year	<i>Bloomberg</i>
Mexico	Policy Rate	–	<i>Bloomberg</i>
Russia	Money Market Rate	–	<i>IMF IFS</i>
South Africa	<i>JIBAR</i>	12-month	South African Reserve Bank
Thailand	Policy Rate	–	<i>Bloomberg</i>

Notes. *EMMI* European Money Markets Institute. *PRIBOR* Prague Interbank Offered Rate. *BUBOR* Budapest Interbank Offered Rate. *JIBAR* Johannesburg Interbank Average Rate. *IMF IFS* International Monetary Fund, International Financial Statistics.

The only other exception is that Thailand's industrial production comes from Bank of Thailand.

Appendix B

Table 7 reports the series used to construct the exchange rate based instrument.

Table 7: Exchange Rate Instrument Series

Country	Exchange Rate	Tenor	Source
Euro Area	USD/EUR	Spot and 12-month	<i>Bloomberg</i>
Russia	USD/RUB	Spot and 12-month	<i>Bloomberg</i>
South Africa	USD/ZAR	Spot and 12-month	<i>Bloomberg</i>

Notes. *USD* United States dollar. *EUR* European Euro. *RUB* Russian Ruble. *ZAR* South African Rand.

Appendix C

An issue raised about high-frequency monetary policy shocks is that they may be predictable from publicly available information, calling into question their exogeneity as instruments. This issue is related to the “information effect”: if markets interpret a surprise rate hike as signaling that the central bank possesses private knowledge of stronger-than-expected fundamentals, then asset price movements around announcements conflate the policy shock with information revelation (see, for instance, Nakamura and Steinsson 2018 and Jarociński and Karadi 2020).

Bauer and Swanson (2023a) provide an alternative explanation. They argue that what appears to be central bank private information largely reflects the systematic policy response to publicly available macroeconomic data. Building on this, Bauer and Swanson 2023b document that a non-trivial share of high-frequency surprises can be forecasted from pre-meeting data, and recommend orthogonalizing surprises with respect to such data before using them as instruments.

To formalize, recall from Section 2 that the proxy z_t must satisfy $\mathbb{E}[z_t \epsilon_{ip,t}] = \mathbb{E}[z_t \epsilon_{p,t}] = 0$. The raw shock can be decomposed as

$$\Delta r_t = \mathbb{E}[\Delta r_t \mid \mathbf{X}_{t-1}] + \varepsilon_t$$

where the first term is the central bank’s systematic reaction to publicly available conditions and ε_t is the exogenous policy innovation. If $\mathbf{X}_{t-1}$ contains variables correlated with non-monetary structural shock (for instance, lagged output growth may predict future supply disturbances) then the conditional mean $\mathbb{E}[\Delta r_t \mid \mathbf{X}_{t-1}]$ inherits this correlation. Using Δr_t directly as z_t would therefore violate exogeneity. The orthogonalized residual ε_t eliminates this channel, since by construction $\mathbb{E}[\varepsilon_t \mid \mathbf{X}_{t-1}] = 0$, ensuring that only the unpredictable policy innovation enters the instrument.

Following this approach, Bolhuis et al. (2024) purges the raw shock Δr_t from its predictable component by estimating

$$\Delta r_t = \alpha + \beta \mathbf{X}_{t-1} + \varepsilon_t$$

separately for each country. The vector $\mathbf{X}_{t-1}$ includes: (i) lagged values of the endogenous variables entering the VAR — the interest rate level, log industrial production growth, the unemployment rate and CPI inflation — which capture the domestic economic conditions to which the central bank may be responding systematically; and (ii) global controls, including the VIX index, changes in oil prices and the U.S. federal funds rate, which account for global financial conditions that could simultaneously drive both domestic monetary policy decisions and exchange rate movements. All controls are dated at $t - 1$ to ensure they belong to the pre-announcement information set.

The residual ε_t is retained as the orthogonalized monetary policy shock and used as the interest rate based external instrument z_t . By construction, it is uncorrelated with publicly available information, addressing the concern raised by Bauer and Swanson (2023a) that the raw instrument may capture the central bank’s predictable reaction function rather than an exogenous policy shift.

Appendix D

Bianchi et al. (2023) and Witheridge (2024)'s framework can be adapted to deal with the deviations that have opened in the CIP since the Great Recession. One simply needs to allow for a time-varying λ_t wedge

$$f_{t,t+1} - e_t = r_{t,t+1} - r_{t,t+1}^* - \lambda_t.$$

If this relationship is considered during the narrow window in which the monetary policy announcement takes place, high frequency monetary shocks can still be measured as

$$\Delta f p_{t,t+1} = \Delta r_t$$

by now assuming that not only $r_{t,t+1}^*$ but also λ_t remain constant during the window.

Appendix E

In this section we present both alternative identification and estimation methods and discuss what assumptions they make and how they behave relatively to Proxy SVAR.

Local Projections IV

As good as Proxy SVAR might be for identification, at the end of the day it still relies on the VAR parametrics. Imposing this additional structure over/on the high frequency shocks gives us statistical power that often comes short with high frequency identification, since we are trying to calculate the effect of a narrow time window on large aggregate macroeconomic variables many periods ahead.³⁴ However, the benefits in terms of statistical power comes at the cost of the additional modelling choices that the VAR assumes. Moreover, VAR model misspecification compounds, given that the IRFs are produced by iterating one period ahead the initial coefficients. The opposite stand in this trade-off is to go fully non-parametric by instrumenting Jordà (2005)'s Local Projections with the high frequency shocks

$$y_{c,t+h}^j = \alpha_{c,0}^j + \beta_{c,h}^j \hat{r}_{c,t}^j + \gamma_{c,h}^j \mathbf{x}_{c,t}^j + \xi_{c,t+h}^j \quad \forall j \in J, c \in C, \forall h \in [0, H]$$

where $y_{c,t+h}^j$ is outcome j for country c at period $t + h$, $\alpha_{c,0}^j$ is a constant associated with outcome j and country c , $\hat{r}_{c,t}^j$ is the prediction coming out from the first stage on the high frequency shock for country c at period t and $\mathbf{x}_t^j$ is a vector of controls associated with outcome j for country c at period t and $\xi_{c,t+h}^j$ are error terms.

Instrumenting Local Projections with the high frequency shocks instead of directly running Local Projections on the shocks tames the aforementioned statistical power concerns and deals with potential measurement error issues. Note that the similarity of the results give us additional confidence on the VAR modelling choices made above, given the aforementioned discussion between parametrics and non-parametrics.

Proxy SVAR Local Projections

An intermediate step in the above bias-variance tradeoff between Proxy SVAR and Local Projections IV is to run Jordà (2005)'s Local Projections on monetary shocks identified from the Proxy SVAR, $\hat{\epsilon}_{1,t}$. This approach exploits the VAR structure only for identification of the shock but does not rely on the parametrics of the model for estimation. More particularly/specifically, what we do is to run the following set of regressions

$$y_{c,t+h}^j = \alpha_{c,0}^j + \beta_{c,h}^j \hat{\epsilon}_{c,t}^j + \gamma_{c,h}^j \mathbf{x}_{c,t}^j + \xi_{c,t+h}^j \quad \forall j \in J, c \in C, \forall h \in [0, H]$$

where $y_{c,t+h}^j$ is outcome j for country c at period $t + h$, $\alpha_{c,0}^j$ is a constant associated with outcome j and country c , $\hat{\epsilon}_{c,t}^j = \hat{\epsilon}_{1,t}^c$ is the high frequency shock identified from the Proxy SVAR for country c at period t and $\mathbf{x}_t^j$ is a vector of controls associated with outcome j for country c at period t and $\xi_{c,t+h}^j$ are error terms.

³⁴ Another gain that Proxy SVAR may have relatively to Local Projections is that one can get additional statistical power by relying on the VAR structure to go further into the past that what the instrument allows for, if the outcome variable sample is larger than the proxy sample. Given the particular nature of our proxy, which exploits deviations from the CIP after the Great Recession, we considered it was conceptually appropriated to only rely on the span of the external instrument sample.

Bayesian Local Projections

Another compromise between bias and variance is Bayesian Local Projections, whose IRF responses posterior means Ferreira et al. (2023) showed to span the space between VAR IRF responses (B_{VAR}^h) and Local Projections ($B_{\text{BLP}}^{(h)}$)

$$B_{\text{BLP}}^{(h)} \propto \left(X'X + \left(\Omega_0^{(h)}(\gamma) \right)^{-1} \right)^{-1} \left(X'Y^{(h)} + \left(\Omega_0^{(h)}(\gamma) \right)^{-1} B_{\text{VAR}}^h \right)$$

The intuition behind such a result is that when $h = 1$, Local Projections collapses to an order one VAR. Therefore, the VAR structure can be seen as an informative prior for Bayesian estimated Local Projections. But, when $h > 1$, now the posterior coming out of the VAR-informed Local Projection serves as a prior. Therefore, as the horizon grows large, Bayesian Local Projections free itself from the VAR structure and deviates from it when necessary.³⁵

Bayesian Proxy SVAR

For the sake of completeness, we present results from Bayesian estimated Proxy SVAR (as opposed to OLS estimated Proxy SVAR). Parametrization is kept as in Ferreira et al. (2023).

³⁵ Bayesian methods require additional parametrization that we keep as in Ferreira et al. (2023).

Appendix F

Bootstrap Test for IRF Estimation Uncertainty. The cross-country regressions in Table 2 use cumulative impulse response functions as the dependent variable. Since these are themselves estimated objects, the OLS standard errors reported in the table do not account for the estimation uncertainty in the IRFs. This appendix presents a simulation exercise to verify that the inverted U-shape relationship between hand-to-mouth shares and monetary policy effectiveness is indeed statistically significant.

The procedure is as follows. For each country c , the Proxy SVAR delivers a point estimate $\hat{\theta}_h^c$ and 90% confidence bands $[\hat{\theta}_h^{c,L}, \hat{\theta}_h^{c,U}]$ at each horizon $h = 0, \dots, 36$. The cumulative response and its bounds are

$$\hat{C}^c = \sum_{h=0}^{36} \hat{\theta}_h^c, \quad \hat{C}^{c,L} = \sum_{h=0}^{36} \hat{\theta}_h^{c,L}, \quad \hat{C}^{c,U} = \sum_{h=0}^{36} \hat{\theta}_h^{c,U}.$$

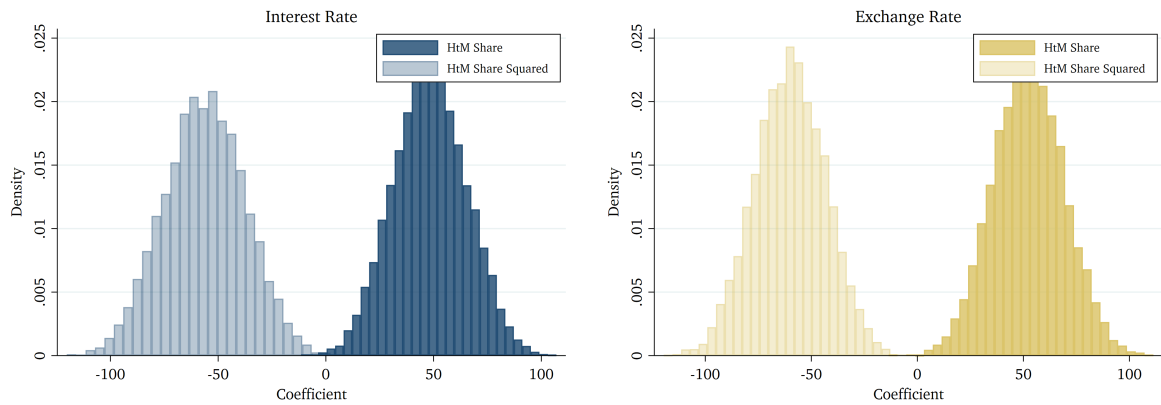
For $b = 1, \dots, 10,000$, draw

$$\tilde{C}^{c,b} \sim \mathcal{U}(\hat{C}^{c,L}, \hat{C}^{c,U})$$

independently across countries and re-estimate the regression.³⁶

Results. Figure 14 displays the distribution of coefficients for both instrument types. For the specification in column 2 of Table 2, the hump shape is preserved in 99.91% of the draws. The median bootstrap coefficient on HtM is 52.9 with a 95% simulation interval of [17.4, 88.8], and the median coefficient on HtM Sq. is -60.7 with interval [-100.0, -21.8]. For column 4's specification, the hump shape is preserved in 99.97% of draws, with median coefficients of 52.0 [19.7, 84.9] and -59.6 [-92.2, -27.6].

Figure 14: Bootstrap Distribution of Regression Coefficients



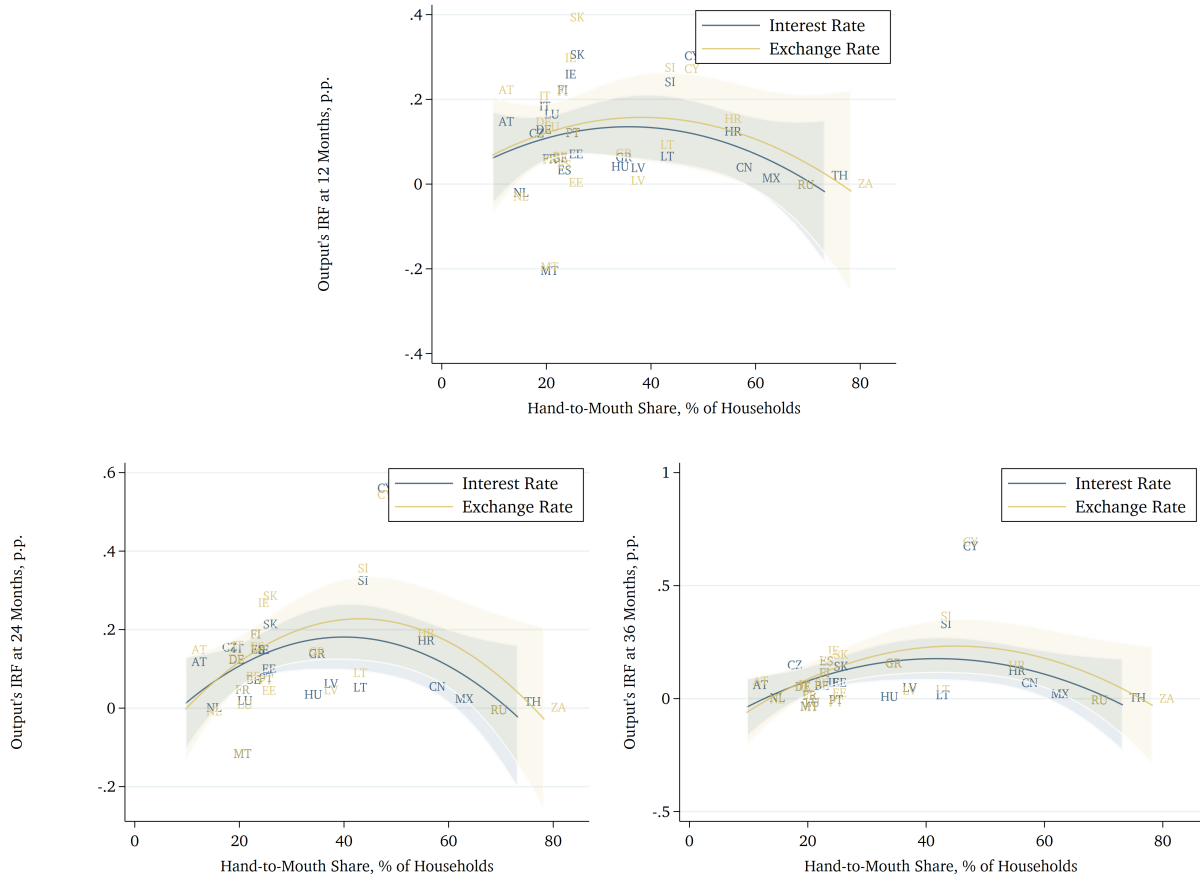
Notes. Distribution of OLS coefficients from 10,000 bootstrap iterations. For each of them, a random draw from each country's cumulative IRF is done from within its 90% Proxy SVAR confidence band. The cross-country quadratic regression is then re-estimated with this draw as the outcome variable.

³⁶ Note that summing across horizons actually *overstates* the uncertainty in $\hat{C}^c$ because it ignores the positive correlation across horizons, making this a conservative test.

Appendix G

Figure 4 plots the cumulative sum of impulse response functions against hand-to-mouth shares, aggregating the full 36-month path into a single measure for each country. This captured how much output was gained or over the whole shock response. Figure 15 instead plots the *level* of the impulse response at $h = 12$, $h = 24$ and $h = 36$ months. This reflects how larger output is relatively to what it would have been without the shock. The inverted U-shape is visible at all three horizons and under both identification strategies, suggesting that the hump-shaped relationship between shocks responses and development is not an artifact of the cumulative aggregation.

Figure 15: Output IRFs at Individual Horizons and Hand-to-Mouth Shares



Notes. Lines are OLS estimated quadratic relationships between the impulse response of industrial production (in percentage points) to a monetary shock at $h = 12$, $h = 24$ and $h = 36$ months and hand-to-mouth shares. A household is classified as such if its net liquid wealth is non-negative but does not exceed one-half of monthly labor income. Shaded areas are 95% confidence bands. Countries with first-stage F-statistics below 10 are excluded for each instrument.

Appendix H

Following the decomposition in Holm et al. (2021), I separate the hand-to-mouth consumption differential into *direct* and *indirect* effects. Direct effects capture channels that operate holding nonfinancial income constant. Indirect effects operate through general equilibrium changes: as monetary policy affects aggregate demand, with resulting income changes feed into consumption. Because hand-to-mouth households have higher MPCs, the indirect income channel disproportionately affects their consumption.

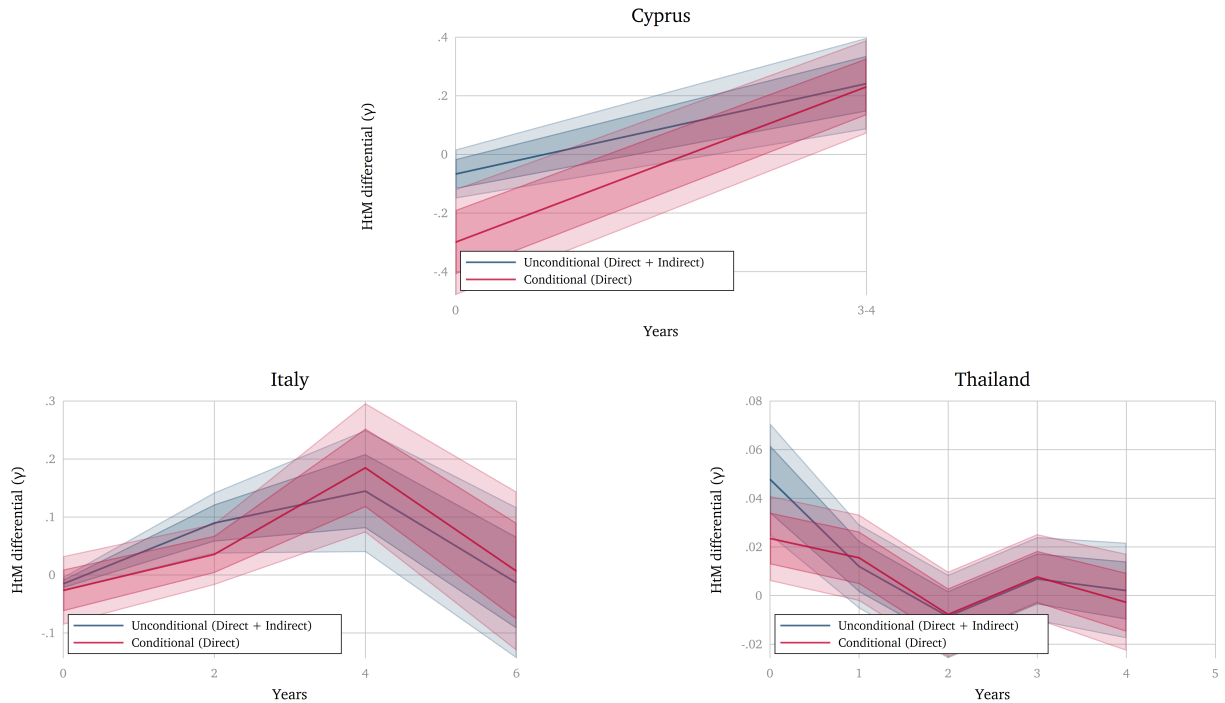
To implement this decomposition, I augment equation 2

$$\log c_{i,t+h} = \alpha_i + \beta_h \hat{r}_t + \gamma_h \left(r_t \times \widehat{\text{HtM}}_i \right) + \delta \log c_{i,t-1} + \theta_{0,h} \log y_{i,t} + \theta_{1,h} \log y_{i,t+1} + \varepsilon_{i,t+h}, \quad (3)$$

where $y_{i,t}$ and $y_{i,t+1}$ denote log labor income at periods t and $t+1$, respectively. Now the estimated $\hat{\gamma}_h$ captures only the *direct* component of the hand-to-mouth differential—the part that is not mediated by income changes.

Figure 16 presents the results. Across all three countries, the hand-to-mouth differential is attenuated when income is held constant, indicating that a significant portion of the heterogeneous consumption response operates through the indirect income channel. Hand-to-mouth households respond more strongly to monetary policy not only because of their direct interest rate exposure, but importantly because they cannot smooth the general equilibrium income changes that monetary policy induces.

Figure 16: Direct vs. Indirect Effects: HtM Differential Controlling for Income



Notes. Navy lines show the baseline hand-to-mouth differential $\hat{\gamma}_h$ from equation 2. Cranberry lines show $\hat{\gamma}_h$ from equation 3, which additionally controls for contemporaneous and one-period-ahead log labor income. Household fixed effects and lagged log consumption (Italy and Thailand) included in both specifications. Standard errors clustered at the household level. Shaded areas are 68% and 90% confidence bands.

Appendix I

The equilibrium of the model (in sequence notation) is given by goods market clearing

$$\mathbf{Y} = \mathcal{C}(\mathbf{r}, \mathbf{Y})$$

where $\mathcal{C}$ is a vector in which each entry is the result of the aggregate consumption function mapping current and future realizations of r and Y to present consumption. For one element of the sequence $\mathbf{Y}$, goods market clearing can be written as

$$Y_t = \mathcal{C}_t(\{r_s, Y_s\})$$

To which we can feed an small shock $\{dr_t\}$

$$dY_t = \sum_{s=0}^{\infty} \frac{\partial \mathcal{C}_t}{\partial r_s} dr_s + \sum_{s=0}^{\infty} \frac{\partial \mathcal{C}_t}{\partial Y_s} dY_s$$

That can be taken back to sequence notation as

$$d\mathbf{Y} = \mathbf{M}^r d\mathbf{r}^{\text{ante}} + \mathbf{M} d\mathbf{Y}$$

Where $\mathbf{M}^r$ and $\mathbf{M}$ are Jacobians, for instance, the aforementioned one $\mathcal{J}^{\mathbf{C}, \mathbf{Y}}$

$$M_{t,s} \equiv \frac{\partial \mathcal{C}_t}{\partial Y_s} = \mathcal{J}_{t,s}^{\mathbf{C}, \mathbf{Y}}$$

In order to solve the fixed point problem, rewrite as

$$(\mathbf{I} - \mathbf{M})d\mathbf{Y} = \mathbf{M}^r d\mathbf{r}$$

where $(\mathbf{I} - \mathbf{M})$ is not invertible but Auclert et al. (2018) showed that a unique solution is

$$d\mathbf{Y} = \mathcal{M} \mathbf{M}^r d\mathbf{r}$$

where $\mathcal{M} \equiv (\mathbf{K}(\mathbf{I} - \mathbf{M}))^{-1} \mathbf{K}$ if and only if $\mathbf{K}(\mathbf{I} - \mathbf{M})$ is invertible.³⁷

³⁷ One possible choice is $\mathbf{K} = -\sum_{t=1}^{\infty} (1+r)^{-t} \mathbf{F}^t$, where $\mathbf{F}$ is the forward operator matrix.

Appendix J

As in the main fact, maybe a quadratic specification better describes the relationship between average excess consumption and hand-to-mouth shares. Table 8 demonstrates that it is not the case. Column 1 confirms that Figure 9's left panel's linear correlation is indeed statistically significant. Column 2 accommodates a quadratic term, which is marginally statistically significant with the correct negative sign. Nevertheless, once one includes in Column 3 the same *WBSI* controls as in section 2's Table 2, both terms become statistically not significant. One could argue that this is due to having too few observations. But, if we drop the quadratic term, the linear one stays with the correct sign and is statistically significant despite controlling for other cross-country variables.

Table 8: Marginal Propensities to Consume and Hand-to-Mouth Shares

	(1)	(2)	(3)	(4)
	Exc. Cons.	Exc. Cons.	Exc. Cons.	Exc. Cons.
HtM Share	0.447** (0.186)	2.697** (1.149)	1.478 (1.030)	2.013*** (0.574)
HtM Share Squared		-2.573* (1.303)	0.788 (1.188)	
Observations	15	15	15	15
R^2	0.195	0.366	0.813	0.806
Controls	No	No	Yes	Yes

Notes. Results of OLS estimated cross-country quadratic relationships between marginal propensity to consume and development. The former is given by average excess consumption estimates compiled in Havranek and Sokolova (2020). The latter is measured as hand-to-mouth shares, where a household is classified as such if its net liquid wealth is larger or equal than zero but smaller or equal than one-half their monthly labor income. Controls from *WBSI* include per capita GDP, share of exports and share of savings. All regressions include a constant. Robust standard errors in parentheses.

Appendix K

Two-asset HANK model with capital accumulation subject to an adjustment cost.

Households. Households allocate their savings between liquid assets b_{it} and illiquid assets a_{it} subject to a convex portfolio adjustment cost $\Phi_t(a_{it}, a_{it-1})$. Hours worked are the same for all households $n_{it} = N_t$. Bellman equation then is

$$\begin{aligned} V_t(e_{it}, b_{it-1}, a_{it-1}) = \max_{c_{it}, b_{it}, a_{it}} & \left\{ \frac{c_{it}^{1-\sigma}}{1-\sigma} - \varphi \frac{N_t^{1+v}}{1+v} + \beta \mathbb{E}_t[V_{t+1}(e_{it+1}, b_{it}, a_{it})] \right\} \\ \text{s.t. } c_{it} + a_{it} + b_{it} = & (1 - \tau_t) w_t N_t e_{it} + (1 + r_t^a) a_{it-1} + (1 + r_t^b) b_{it-1} - \Phi_t(a_{it}, a_{it-1}) \\ a_{it} \geq 0, \quad b_{it} \geq & \underline{b}. \end{aligned}$$

The adjustment cost function is given by

$$\Phi_t(a_{it}, a_{it-1}) = \frac{\chi_1}{\chi_2} \left| \frac{a_{it} - (1 + r_t^a) a_{it-1}}{(1 + r_t^a) a_{it-1} + \chi_0} \right|^{\chi_2} [(1 + r_t^a) a_{it-1} + \chi_0]$$

Financial Intermediary. A representative financial intermediary takes households' liquid and illiquid deposits and invests them in firm equity p_t and government bonds B_t^g . Illiquid assets can be converted into liquid assets at cost $\omega \int b_{it} di$. The no-arbitrage conditions imposes that the ex-ante return $\mathbb{E}_t[1 + r_{t+1}]$ equals the expected returns on nominal government bonds and on equity. Since the financial intermediary is competitive, it passes these returns to households, who need to pay an intermediation cost

$$\mathbb{E}_t[1 + r_{t+1}] = \frac{1 + i_t}{\mathbb{E}_t[1 + \pi_{t+1}]} = \frac{\mathbb{E}_t[d_{t+1} + p_{t+1}]}{p_t} = \mathbb{E}_t[1 + r_{t+1}^a] = \mathbb{E}_t[1 + r_{t+1}^b] + \omega$$

The ex-post returns r_t, r_t^a, r_t^b are subject to surprise inflation and capital gains. Assuming that capital gains accrue to the illiquid account, we have $1 + r_t = (1 + i_{t-1}) / (1 + \pi_t) = 1 + r_t^b + \omega$ and

$$1 + r_t^a = \Theta_p \left(\frac{d_t + p_t}{p_{t-1}} \right) + (1 - \Theta_p) (1 + r_t)$$

where Θ_p denotes the share of equity in the illiquid portfolio.

Firms. Intermediate goods firms have a Cobb Douglas production function $y_{jt} = F(k_{jt-1}, n_{jt}) \equiv k_{jt-1}^\alpha n_{jt}^{1-\alpha}$. Firms choose capital subject to a quadratic adjustment costs $\zeta \left(\frac{k_{jt}}{k_{jt-1}} \right) k_{jt-1}$ with $\zeta(x) \equiv x - (1 - \delta) + \frac{1}{2\delta\epsilon_I}(x - 1)^2$, where $\delta > 0$ is depreciation and $\epsilon_I > 0$. The Phillips curve is given by $mc_t = w_t / F_N(K_{t-1}, N_t)$. Aggregate investment by $I_t = K_t - (1 - \delta)K_{t-1} + \zeta \left(\frac{K_t}{K_{t-1}} \right) K_{t-1}$. Dividends equal output net of investment, labor costs, and price adjustment costs $d_t = Y_t - w_t N_t - I_t - \psi_t$. Tobin's Q and capital follow

$$\begin{aligned} Q_t &= 1 + \frac{1}{\delta\epsilon_I} \frac{K_t - K_{t-1}}{K_{t-1}} \\ (1 + r_{t+1}) Q_t &= \alpha \frac{Y_{t+1}}{K_t} mc_{t+1} - \left[\frac{K_{t+1}}{K_t} - (1 - \delta) + \frac{1}{2\delta\epsilon_I} \left(\frac{K_{t+1} - K_t}{K_t} \right)^2 \right] + \frac{K_{t+1}}{K_t} Q_{t+1} \end{aligned}$$

Unions. A competitive union aggregates labor with constant elasticity of substitution $\mu_w/(\mu_w - 1) > 1$. The wage for each labor type is set by a each labor union. Unions optimize by setting wages to maximize the average utility of households, taking as given their consumption-savings decisions. Setting a nominal wage W_{kt} incurs quadratic adjustment cost $\psi_t^w(W_{kt}, W_{kt-1}) = \frac{\mu_w}{\mu_w - 1} \frac{1}{2\kappa_w} [\log(W_{kt}/W_{kt-1})]^2$ in utils. Aggregate wage inflation $1 + \pi_t^w = (1 + \pi_t) w_t/w_{t-1}$ evolves according to the Phillips curve

$$\log(1 + \pi_t^w) = \kappa_w \left(\varphi N_t^{1+v} - \frac{(1 - \tau_t) w_t N_t}{\mu_w} \int e_{it} c_{it}^{-\sigma} di \right) + \beta \log(1 + \pi_{t+1}^w)$$

Policy. τ_t is a proportional tax on labor and budget balance requires $\tau_t w_t N_t = r_t B^g + G_t$.

Market Clearing. Good markets clear $Y_t = \int c_{it} di + G_t + I_t + \psi_t + \omega \int b_{it-1} di + \int \Phi_t(a_{it}, a_{it-1}) di$ and assets market too $p_t + B^g = \int a_{it} + b_{it} di$.

Table 9: Calibration of Two-Asset HANK Model

Parameter		Value	Target
<i>A. Households</i>			
β	Discount factor	0.976	$r = 0.0125$
σ	Inverse IES	2	
χ_0	Portfolio adj. cost pivot	0.25	
χ_1	Portfolio adj. cost scale	6.416	$B = 1.04Y$
χ_2	Portfolio adj. cost curvature	2	
$\underline{b}$	Borrowing constraint	0	
ρ_e	Autocorrelation of earnings	0.966	
σ_e	Cross-sectional std of log earnings	0.92	
<i>B. Labor unions</i>			
φ	Disutility of labor	2.073	$N = 1$
ν	Inverse Frisch elasticity	1	
μ_w	Steady state wage markup	1.1	
κ_w	Slope of wage Phillips curve	0.1	
<i>C. Firms</i>			
Z	TFP	0.468	$Y = 1$
α	Capital share	0.33	$K = 10Y$
μ_p	Steady-state markup	1.015	$p + B^g = 14Y$
δ	Depreciation	0.02	
κ_p	Slope of price Phillips curve	0.1	
<i>D. Financial intermediary</i>			
ω	Liquidity premium	0.005	
<i>E. Policy</i>			
τ	Labor tax	0.356	budget balance
G	Government spending	0.2	
B^g	Bond supply	2.8	
ϕ	Taylor rule coefficient	1.5	
ϕ_y	Taylor rule coefficient on output	0	
<i>F. Discretization</i>			
n_e	Points in Markov chain for e	3	
n_b	Points on liquid asset grid	50	
n_a	Points on illiquid asset grid	70	